DISTANCE LEARNING CENTRE

AHMADU BELLO UNIVERSITY

ZARIA, NIGERIA

COURSE MATERIAL

FOR

COURSE CODE &TITLE: LIBS 880: INFORMATION SYSTEM ANALYSIS AND DESIGN

PROGRAMME TITLE: MASTERS IN INFORMATION MANAGEMENT

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QUOTE

Sometime in the future, science will be able to create realities that we cannot even begin to imagine. As we evolve, we will be able to construct other information systems that correspond to other realities based on logic completely different from ours and not based on space and time…

Robert lanza

TABLE OF CONTENT

Title Page 1
Copyright Page 3
Course Writers/Development Team 4
Table of Content 6

COURSE STUDY GUIDE.. .. ..
Preamble .. ..
i. Course Information 8
ii. Course Introduction and Description 8
iii. Course Prerequisites 8
iv. Course Learning Resources 9
v. Course Objectives and Outcomes 9
vi. Activities to Meet Course Objectives 9
vii. Time (To Complete Syllabus/Course) 10
viii. Grading Criteria and Scale 10
ix. OER Resources 11
x. ABU DLC Academic Calendar 13
xi. Course Structure and Outline 14
xii. STUDY MODULES 21
1.0 Module 1: Information Systems 22
Study Session 1: Definition of System 22
Study Session 2: Elements of a system 30
Study Session 3: Information system 38
Study Session 4: Types of Information system 49

2.0 Module 2: Information System Analysis 57
Study Session 1: System Analysis 57
Study Session 2: Types of System Analysis 66
Study Session 3: Who is a System Analyst 73
Study Session 4: Roles and Skills of System Analysts 79

3.0 Module 3:Information System Design and Development Lifecycle 85
Study Session 1: Definition of System Design 85
Study Session 2: System Design tools 95
Study Session 3: System Development Lifecycle (SDLC) 101
Study Session 4: System Development Models 110
XIII. Glossary .. ..

Oftware

i. COURSE INFORMATION

Course Code :LIBS 880

Course Title: Information System Analysis and Design

Credit Units: 2

Semester: Second

ii. COURSE INTRODUCTION AND DESCRIPTION

This course will introduce you to the concepts, theories and techniques of **information systems analysis and design**. The main goal of systems analysis and design is to improve **organizational systems**, typically through applying software that can help employees accomplish key business tasks more easily and efficiently.

iii. COURSE PREREQUISITES

You are welcome to LIBS 880, this is a 2credit unit second semester course. I will be your guide in the span of this course, you should feel free to ask questions when in a form of difficulties relating this course, and my door is always open.

You should note that although this course has no subject pre-requisite, you are expected to have:

1. Satisfactory level of English proficiency
2. Basic Computer Operations proficiency
3. Online interaction proficiency
4. Web 2.0 and Social media interactive skills

iv. COURSE LEARNING RESOURCES

i. Course Textbooks

Dennis, A., Wixom, B. H., & Roth, R. M. (2014).*Systems analysis and design*. John wiley& sons.

Dennis, A., Wixom, B. H., &Tegarden, D. (2015).*Systems analysis and design: An object-oriented approach with UML*. John wiley& sons.

https://www.youtube.com/watch?v=XlcolUHMnh0

v. ACTIVITIES TO MEET COURSE OBJECTIVES

Specifically, this course shall comprise of the following activities:

1. Studying courseware
2. Listening to course audios
3. Watching relevant course videos
4. Field activities, industrial attachment or internship, laboratory or studio work (whichever is applicable)
5. Course assignments (individual and group)
6. Forum discussion participation
7. Tutorials (optional)
8. Semester examinations (CBT and essay based).

vi. TIME (TO COMPLETE SYLABUS/COURSE)

To cope with this course, you would be expected to commit a minimum of 3 hours Weekly for the Course.

vii. GRADING CRITERIA AND SCALE

Grading Criteria

A. Formative assessment

Grades will be based on the following:

Individual assignments/test (CA 1,2 etc)	20
Group assignments (GCA 1, 2 etc)	10
Discussions/Quizzes/Out of class engagements etc	10

B. Summative assessment (Semester examination)

CBT based	30
Essay based	30
TOTAL	**100%**

C. Grading Scale:

A = 70-100

B = 60 – 69

C = 50 - 59

D = 45-49

F = 0-44

D. Feedback

Courseware based:

1. In-text questions and answers (answers preceding references)
2. Self-assessment questions and answers (answers preceding references)

Tutor based:

1. Discussion Forum tutor input
2. Graded Continuous assessments

Student based:

1. Online programme assessment (administration, learning resource, deployment, and assessment).

viii. LINKS TO OPEN EDUCATION RESOURCES

OSS Watch provides tips for selecting open source, or for procuring free or open software.

SchoolForge and SourceForge are good places to find, create, and publish open software. SourceForge, for one, has millions of downloads each day.

Open Source Education Foundation and Open Source Initiative, and other organisation like these, help disseminate knowledge.

Creative Commons has a number of open projects from Khan Academy to Curriki where teachers and parents can find educational materials for children or learn about Creative Commons licenses. Also, they recently launched the School of Open that offers courses on the meaning, application, and impact of "openness."

Numerous open or open educational resource databases and search engines exist. Some examples include:

- OEDb: over 10,000 free courses from universities as well as reviews of colleges and rankings of college degree programmes
- Open Tapestry: over 100,000 open licensed online learning resources for an academic and general audience
- OER Commons: over 40,000 open educational resources from elementary school through to higher education; many of the elementary, middle, and high school resources are aligned to the Common Core State Standards
- Open Content: a blog, definition, and game of open source as well as a friendly search engine for open educational resources from MIT, Stanford, and other universities with subject and description listings
- Academic Earth: over 1,500 video lectures from MIT, Stanford, Berkeley, Harvard, Princeton, and Yale

- JISC: Joint Information Systems Committee works on behalf of UK higher education and is involved in many open resources and open projects including digitising British newspapers from 1620-1900!

Other sources for open education resources

Universities

- The University of Cambridge's guide on Open Educational Resources for Teacher Education (ORBIT)
- OpenLearn from Open University in the UK

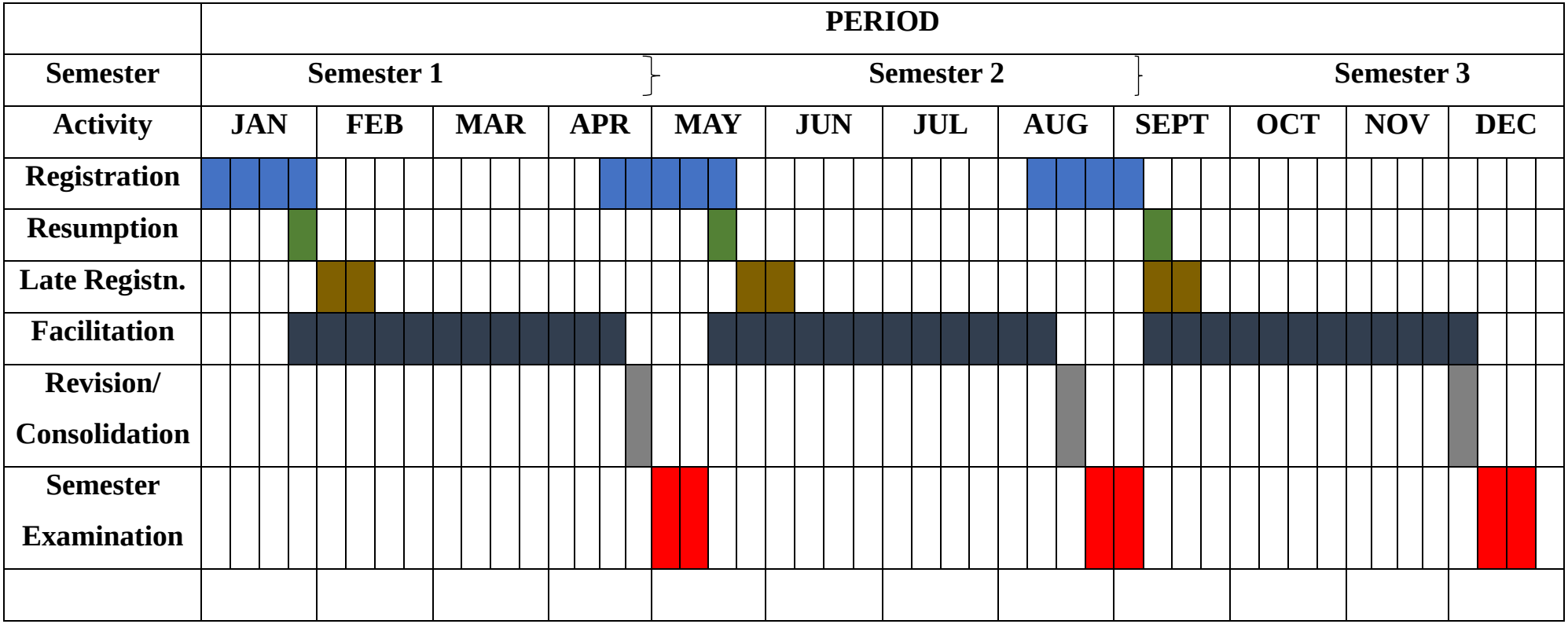
Global

- Unesco's searchable open database is a portal to worldwide courses and research initiatives
- African Virtual University (http://oer.avu.org/) has numerous modules on subjects in English, French, and Portuguese
- https://code.google.com/p/course-builder/ is Google's open source software that is designed to let anyone create online education courses
- Global Voices (http://globalvoicesonline.org/) is an international community of bloggers who report on blogs and citizen media from around the world, including on open source and open educational resources

Individuals (which include OERs)

- Librarian Chick: everything from books to quizzes and videos here, includes directories on open source and open educational resources
- K-12 Tech Tools: OERs, from art to special education
- Web 2.0: Cool Tools for Schools: audio and video tools
- Web 2.0 Guru: animation and various collections of free open source software
- Livebinders: search, create, or organise digital information binders by age, grade, or subject (why re-invent the wheel?)

ix. ABU DLC ACADEMIC CALENDAR/PLANNER



N.B: - All Sessions commence in January

- 1 Week break between Semesters and 6 Weeks vocation at end of session.

- Semester 3 is OPTIONAL (Fast-tracking, making up carry-overs & deferments)

x. COURSE STUDY GUIDE AND OUTLINE

Course Study Guide

WEEK	MODULE	STUDY SESSION	ACTIVITY
Week 1	**STUDY MODULE 1**	**Study Session 1:** Definition of System ***Pp. 22***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site (https://bit.ly/2BlQP6D) 5. View referred Animation (Address/Site https://bit.ly/2Lqi1Zj)
Week 2		**Study Session 2** Elements of a system ***Pp. 30***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site (https://bit.ly/2GjTB04) 5. View referred Animation (Address/Site https://bit.ly/2LopDeU)
Week 3		**Study Session 3** Information systems ***Pp. 38***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/2GfBUyV) 5. View referred Animation (Address/Site https://bit.ly/2xdQdip)
Week 4		**Study Session 4** Types of Information system ***Pp. 49***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/sitehttps://bit.ly/2WJwcu2 https://bit.ly/2DkN0yz https://bit.ly/2BoUbFW)

			5. View referred Animation (Address/Site https://bit.ly/2DRCobH)
Week 5	**STUDY MODULE 2**	**Study Session 1** System Analysis ***Pp. 57***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/2Gt1dwN) 5. View referred Animation (Address/Site https://bit.ly/2WpuRId)
Week 6		**Study Session 2** Types of System Analysis ***Pp. 66***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/2Gt1dwN) 5. View referred Animation (Address/Site https://bit.ly/2TGWYAs)
Week 7		**Study Session 3** Who is a System Analyst ***Pp. 73***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/2Gt1dwN) 5. View referred Animation (Address/Site https://bit.ly/2WpuRId)
Week 8		**Study Session 4** Skills of System Analysts ***Pp. 79***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/1IunfcH) 5. View referred Animation (Address/Site https://bit.ly/2N26QDf)
			1. Read Courseware for the corresponding Study Session.

Week 9	**STUDY MODULE 3**	**Study Session 1** Definition of System Design ***Pp.85***	2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/2MWJcYS) 5. View referred Animation (Address/Site https://bit.ly/2V7QAmQ)
Week 10		**Study Session 2** System Design tools ***Pp. 95***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/2E4KXjH) 5. View referred Animation (Address/Site https://bit.ly/2IXR39i)
Week 11		**Study Session 3** System Development Lifecycle (SDLC) ***Pp. 101***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/ZcpJJP) 5. View referred Animation (Address/Site https://bit.ly/2txfToA)
Week 12		**Study Session 4** System Development Models ***Pp. 110***	1. Read Courseware for the corresponding Study Session. 2. View the Video(s) on this Study Session 3. Listen to the Audio on this Study Session 4. View any other Video/U-tube (address/site https://bit.ly/2RWZfqj https://bit.ly/2SKAUsm) 5. View referred Animation (Address/Site https://bit.ly/2vFK9wS)
Week 13		**REVISION/TUTORIALS (On Campus or Online)& CONSOLIDATION WEEK**	

Week 14& 15		**SEMESTER EXAMINATION**

Course Outline

MODULE 1: Information Systems

Study Session 1: Definition of System

Study Session 2: Elements of a system

Study Session 3: Information system

Study Session 4: Types of Information system

MODULE 2: Information Systems Analysis

Study Session 1: System Analysis

Study Session 2: Types of System Analysis

Study Session 3: Who is a System Analyst?

Study Session 4: Skills of System Analysts

MODULE 3: Information Systems Design and Development Lifecycle

Study Session 1: Definition of System Design

Study Session 2: System Design tools

Study Session 3: System Development Lifecycle (SDLC)

Study Session 4: System Development Models

xii. STUDY MODULES

MODULE 1: Information Systems

Contents:

Study Session 1: Definition of System

Study Session 2: Elements of systems

Study Session 3: Information system

Study Session 4: Types of Information system

STUDY SESSION 1

Definition of System

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

- 2.1-Definition of System
- 2.2- Characteristics of a System
- 2.3- Types of system

3.0Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0Additional Activities (Videos, Animations & Out of Class activities)

6.0References/Further Readings

Introduction:

You are welcome to this study session 1, in this study session, we will discuss and explore the concept of systems, the characteristics of a **system**, the elements of a system and other **system** related concepts.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Define Systems
2. Identify and explain the characteristics of a system
3. Discuss the types of system

2.0 Main Content

2.1 Definition of System

You should have it in mind that, a**system** is an orderly grouping of interdependent components linked together according to a plan to achieve a specific objective. You should note that the word "**system**" is derived from the Greek word '**Systema**' which means an organized relationship among functioning units or components. In Information **Systems** discipline, **systems** are defined as a set of interrelated components that function seamlessly together to generate process and disseminate information.These components are subsystems in their own rights.

Fig 1.1.1: Definition of System
Source: siasfirst.com

Note that, they also function to achieve a particular **system goal**. Examples of **systems** include the Computer system, **telephone system**, library system. In all these examples the **systems** are made up component parts that work together to achieve specific objectives. In the Computer system for example, you have to note that, it is made up of components like the monitor, the system unit, the keyboard and the mouse. Similarly, the **telephone system** (GSM) is made up of the Handset (**phone**), the Mast, and the Mobile switching centre. The **library system** is made up of the technical section, the circulation section, the administrative section and the reserve section.**Systems** are created to solve

problems. One can think of the **systems** approach as an organized way of dealing with a problem. Understanding systems and how they work is critical to understanding **systems analysis and design**. To better understand systems, it is important to identify their characteristics.

In-text Question
What is a system?

Answer
A system is an orderly grouping of interdependent components linked together according to a plan to achieve a specific objective

2.2Characteristics of a System

The following are the characteristics of a **system**:

Organization: This implies structure and order. In the arrangement of component parts of a **system**, there must be order and this gives the **system structure**.

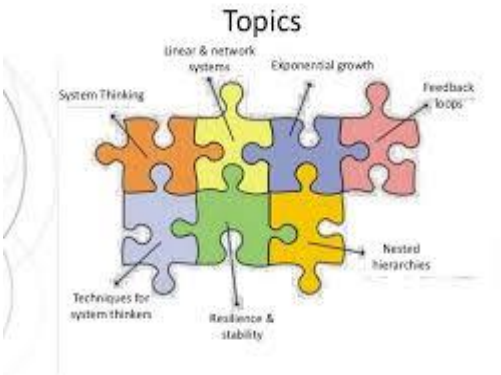


Fig 1.1.2: Characteristics of a System
Source: slideshare.net

For example, your everyday**computer system** is designed around an input device, a central processing unit, an output device and one or more storage units. When linked together they work as a whole **system** for producing information for you. This arrangement portrays the **system–subsystem** relationship, defines the authority structure, specifies the formal flow of communication and formalizes the chain of command.

1. Interaction: It refers to manner in which each component functions with other components of the **system**,you can equally explain this by saying the same way we operate as a family member were everyone plays a different role to archive a common goal. In a **computer system** for example, the central processing unit must interact with the input device to

solve a problem. In turn, the main memory holds programs and data that the arithmetic unit uses for computation. The interrelationship between these components enables the computer to perform.

2. Interdependence: In a **system** all units/parts are dependent on each other. No subsystem can function in isolation because it is dependent on the data (**inputs**) it receives from other subsystems to perform its required tasks. Interdependence is further illustrated by the activities and support of **systems analysts**, programmers, and the operations staff in a computer centre.
3. Integration: The parts of a **system work** together within the system even though each part performs a unique function.You should note that integration is concerned with how a **system** is tied together. It is more than sharing a physical part or location. It means that parts of the **system** work together within the **system** even though each part performs a unique function. Successful integration will typically produce a synergistic effect and greater total impact than if each component works separately.
4. Central Objective: Every System has an objective which may be real or stated. All the components work together to achieve that particular objective.Although a stated objective may be the real objective, it is not uncommon for an organization to state one objective and operate to achieve another. The important point is that users must know the central objective of a computer application early in the analysis for a successful design and conversion.

In-text Question
Discuss any characteristic of a system?

Answer
Every System has an objective which may be real or stated. All the components work together to achieve that particular objective.

2.3 Types of Systems

The systems can be divided into the following types:

Physical or Abstract Systems

Physical **systems** are tangible entities. You can touch and feel them. They may be static or dynamic in nature. For example, your desks and chairs are the physical parts of computer centre which are static. A programmed computer is a dynamic **system** in which programs, data, and applications can change according to the user's needs. Abstract **systems** are non-physical entities or conceptual that may be formulas, representation or model of a real **system**.

Open or Closed Systems

You should try to differentiate them apart. An open **system** must interact with its environment. It receives inputs from and delivers outputs to the outside of the **system**. For example, an **information system** which must adapt to the changing environmental conditions. A closed system does not interact with its environment. Let me give a good example, android is an **open system**,while IOS is a close **system**. It is isolated from environmental influences. A completely closed **system** is rare in reality.

Adaptive and Non-Adaptive System

You should have it back of your mind that, Adaptive **System** responds to the change in the environment in a way to improve their performance and to survive. For example, human beings, animals. Non-Adaptive System is the **system** which does not respond to the environment. For example, machines.

Permanent or Temporary System

Permanent **System** persists for long time. For example, business policies. Temporary **System** is made for specified time and after that they are demolished. For example, A DJ system is set up for a program and it is dissembled after the program.

Natural and Manufactured System

Natural **systems** are created by the nature. For example,your Solar system, seasonal system. Manufactured **System** is the man-made system. For example, Rockets, dams, trains.

Deterministic or Probabilistic System

Deterministic system operates in a predictable manner and the interaction between **system** components is known with certainty. For example, two molecules of hydrogen and one molecule of oxygen makes water. Probabilistic System shows uncertain behaviour. The exact output is not known. For example, Weather forecasting, mail delivery.

Social, Human-Machine, Machine System

Social System is made up of people. For example, your social clubs, societies. In **Human-Machine System**, both human and machines are involved to perform a particular task. For example, Computer programming. Machine **System** is where human interference is neglected. All the tasks are performed by the machine. For example, an autonomous robot.

Man-Made Information Systems

It is an interconnected set of information resources to manage data for particular organisation, under Direct Management Control (DMC). This **system** includes hardware, software, communication, data, and application for producing information according to the need of an organisation.

Man-made information systems are divided into three types:

1. **Formal Information System:** It is based on the flow of information in the form of memos, instructions, etc., from top level to lower levels of management.

2. **Informal Information System**: This is employee-based system which solves the day to day work related problems.
3. **Computer Based System**: This system is directly dependent on the computer for managing business applications. For example, automatic library **system**, railway reservation **system**, banking system, etc.

In-text Question
What type of system is the human being and why?

Answer
The human being is a Adaptive system because it responds to changes in its environment in other to survive

3.0 Conclusion/Summary

This study session provided the student with some of the definitions of a **system**, its characteristics and types. I know you have enjoyed all we have discussed so far.

4.0 Self-Assessment Questions

1. Why do many systems fail in Nigeria?
2. Explain why the central objective and organization characteristics are very important to any system.

6.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=pOzSlT2Chi0&t=57s Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2Lqi1Zj and critique it in the discussion forum

7.0 References/Further Readings

http://www.ddegjust.ac.in/studymaterial/pgdca/ms-04.pdf

Laudon, K.C. and J.P. Laudon (2013). Management information systems*:* managing the digital firm. Boston; London: Pearson, thirteenth edition [ISBN 9780273789970 (pbk)].

Shelly, G. B., & Rosenblatt, H. J. (2012).„System Analysis and Design 9th Edition “.*Shelly Cashman Series*.

STUDY SESSION 2

Elements of aSystem

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1- Elements of a system

2.1.1- Outputs and inputs.

2.1.2- Processor(s).

2.1.3- Control.

2.1.4- Feedback.

2.1.5- Environment.

2.1.6- Boundaries and interface.

2.2- System Related Concepts

3.0Study Session Summary and Conclusion

4.0Self-Assessment Questions

5.0Additional Activities (Videos, Animations & Out of Class activities)

6.0References/Further Readings

Introduction

You are welcome to study session 2, In this session the elements of a **system** are discussed. You will learn the basic elements of a **system** and some **system** related concepts.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Identify the elements of a system
2. Discuss some specific system related concepts

2.0 Main Content

2.1 Elements of a System

In most cases, **systems analysts operate** in a dynamic environment where change is a way of life. The environment may be a business firm, a business application, or a computer system.

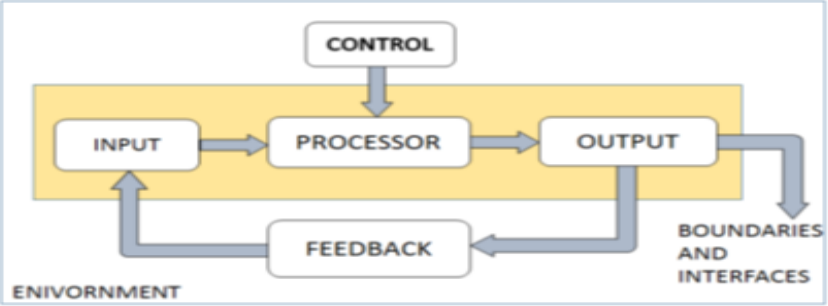


Fig 1.2.1 the following diagram shows the elements of a system

2.1.1 Outputs and Inputs

A major objective of a **system** is to produce an output that has value to its user. Whatever the nature of the output (**goods, services, or information**), it must be in line with the expectations of the intended user. From the above, you will realise that, Inputs are the elements (material, human resources, and information) that enter the system for processing. Output is the outcome of processing. Your **system** feeds on input to produce output in much the same way that a business brings in human, financial, and material resources to produce goods and services. It is important that you point out here that determining the output is a first step in specifying the nature, amount, and regularity of the input needed to operate a system. For example, in **systems** analysis, the first concern is to determine the user's requirements of a proposed computer **system** – that is, specification of the output that the computer is expected to provide for meeting user requirements.

In-text Question
Define Elements of a System

Answer
Systems analysts operate in a dynamic environment where change is a way of life. The environment may be a business firm, a business application, or a computer system.

2.1.2 Processor(s)

The processor (**just like the human brain**) is the element of a **system** that involves the actual transformation of input into output. It is the operational component of a **system**. Processors may modify the input totally or partially, depending on the specifications of the output. This means that as the output specifications change so does the processing. In some cases, input is also modified to enable the processor to handle the transformation.

2.1.3 Control

The control element guides the **system**. It is the decision making subsystem that controls the pattern of activities governing input, processing, and output. In an organisational context, management as a decision making body controls the inflow, handling and outflow of activities that affect the welfare of the business. You should note that, in a computer **system**, the operating system and accompanying software influence the behaviour of the **system**. Output specifications determine what and how much input is needed to keep the **system** in balance.

In **systems analysis**, knowing the attitudes of the individual who controls the area for which a computer is being considered can make a difference between the success and failure of the installation. Management support is required for securing control and supporting the objective of the proposed change.

2.1.4 Feedback

Control in a dynamic **system** is achieved by feedback. Feedback measures output against a standard in some form of **cybernetic** procedure that includes communication and control. Output information is fed back to the input and / or to management (**Controller**) for deliberation. After the output is compared against performance standards, changes can result in the input or processing and consequently, the output. Feedback may be positive or negative, routing or

informational. Positive feedback reinforces the performance of the **system**. It is routine in nature. Negative feedback generally provides the controller with information for action. In **systems analysis**, feedback is important in different ways. During analysis, you may be told that the problems in a given application verify the initial concerns and justify the need for change. Another form of feedback comes after the system is implemented. The user (you) informs the analyst about the performance of the new installation. This feedback often results in enhancements to meet the user's requirements.

2.1.5 Environment

The environment is the "**suprasystem**" within which an organisation operates. It is the source of external elements that impinge on the **system**. In fact, it often determines how a system must function. For example, the organisation's environment, consisting of vendors, competitors, and others, may provide constraints and, consequently, influence the actual performance of the business.

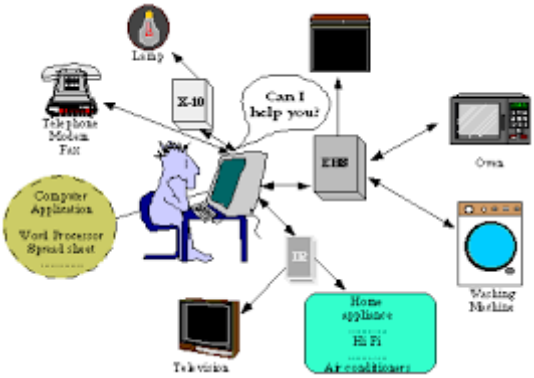


Fig 1.2.2: Environment

Source: ercim.eu

2.1.6 Boundaries and interface

A **system** should be defined by its boundaries, just like in everyday day to day activities you have boundaries, the limits that identify its components, processes and interrelationship when it interfaces with another **system**. For example, a teller **system** in a commercial bank is restricted to the deposits, withdrawals and related activities of customers checking and savings accounts. It may exclude mortgage foreclosures, trust activities, and the likes.

Each **system** has boundaries that determine its sphere of influence and control. For example, in an integrated banking – wide computer **system design**, a customer who has a mortgage and a checking account with the same bank may write a check through the "**teller system**" to pay the premium that is later processed by the "**mortgage loan system**." Recently, **system design** has been successful in allowing the automatic transfer of funds form a bank account to pay bills and other obligations to creditors, regardless of distance or location. This means that in systems analysis, knowledge of the boundaries of a given **system** is crucial in determining the nature of its interface with other systems for successful design.

In-text Question
Why is the Environment component of a system seen as a "Suprasystem" within which an organization operates?

Answer
It is seen as the "Suprasystem" because it constitutes the external sources of elements that influence the performance of the system.

2.2 System Related Concepts

Systems analysts need to know several other important **systems** concepts:

1. Decomposition
2. Modularity
3. Coupling
4. Cohesion

1. **Decomposition** is the process of breaking down a **system** into its smaller components. These components may themselves be systems (**subsystems**) and can be broken down into their components as well. It results in smaller and less complex pieces that are easier to understand than larger, complicated pieces.

Decomposing a **system** also allows us to focus on one particular part of a **system**, making it easier to think of how to modify that one part independently of the entire **system**. Decomposition is a technique that allows the **systems analyst** to:

a. Break a system into small, manageable, and understandable subsystems
b. Focus attention on one area (**subsystem**) at a time, without interference from other areas
c. Concentrate on the part of the system pertinent to a particular group of users, without confusing users with unnecessary details
d. Build different parts of the system at independent times and have the help of different analysts.

2. **Modularity** is a direct result of decomposition. It refers to dividing a system into chunks or modules of a relatively uniform size. Modules can represent a **system** simply, making it easier to understand and easier to redesign and rebuild. For example, a human being could be decomposed into the head, the body and the legs. Modules for the Human **System** will then include the Nervous System, the Respiratory System, theReproductory System etc. Modularity is grouping some of the parts of a **system** together making them subsystems on their own.

3. **Coupling** means that subsystems are dependent on each other. Subsystems should be as independent as possible. If one subsystem fails and other subsystems are highly dependent on it, the others will either fail themselves or have problems functioning. The Eyes for example sees where the body goes and the legs move the body such that if there is a problem with the eye, moving from one place to the other becomes a problem.

4. **Cohesion** is the extent to which a **subsystem** performs a single function.

In-text Question
State two ways decomposition helps system analysts?

Answer
1. It helps in breaking down the system into manageable parts/subsystems
2. Concentrate on the part of the system pertinent to a particular group of users, without confusing users with unnecessary details

3.0Conclusion/Summary

This study session provided the student with the understanding of elements of a **system** and some **system** related concepts that are used in general **system analysis and design**.

4.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=XlcolUHMnh0and https://www.youtube.com/watch?v=pOzSlT2Chi0&t=57s Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2LopDeU and critique it in the discussion forum

6.0References/Further Readings

Dennis, A., Wixom, B. H., & Roth, R. M. (2014). *Systems analysis and design*. John wiley& sons.

Dennis, A., Wixom, B. H., &Tegarden, D. (2015). *Systems analysis and design: An object-oriented approach with UML*. John wiley& sons.

https://www.youtube.com/watch?v=XlcolUHMnh0

STUDY SESSION 3
Information Systems

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1- Definition of Information System

2.2- Components of an information System

2.2.1- Hardware

2.2.2- Software

2.2.3- Data

2.2.4- Network

2.2.5- People

3.0Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0Additional Activities (Videos, Animations & Out of Class activities)

6.0References/Further Readings

Introduction:

You are welcome to study session 3, you will expose the concept of **information system**. In this session you will learn the definition and the components of an information system.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to

1. Define information systems
2. Identify and discuss the various components of an information system.

2.0 Main Content

2.1 Definition of Information System

It is important for you to note that,An**information system** combines information technology, people, and data to support business requirements. For example, information **systems handle** daily business transactions, improve company productivity, and help managers make sound decisions. Ideally, information reduces uncertainty about a state or event. For example, information that the wind is calm reduces the uncertainty that the boat trip will be pleasant. An information system is the basis for interaction between the user and the analyst.

Fig 1.3.1: Definition of Information System

Source: keyin.ca

It provides instruction, commands and feedback. It determines the nature of the relationships among **decision-makers**. In fact, it may be viewed as a decision centre for personnel at all levels. From this basis, an information system may be defined as a set of devices, procedures and operating systems designed around user-based criteria to produce information and communicate it to the user for planning, control and performance.

In systems analysis, it is important for you to keep in mind that, considering an alternative **system** means improving one or more of these criteria. According to Avison and Fitzgerald (1995) “Information Systems (IS) is a field of research concerned with the effective design, delivery, use and impact of information technology in organisations and society”.

Information system has been defined in terms of two perspectives: one relating to its function; the other relating to its structure. From a functional perspective; an information **system** is a technologically implemented medium for the purpose of recording, storing, and disseminating linguistic expressions as well as for the supporting of inference making. From a structural perspective; an information system consists of a collection of people, processes, data, models, technology and partly formalized language, forming a cohesive structure which serves some organizational purpose or function.

The functional definition has its merits in focusing on what actual users from a conceptual point of view- do with the information system while using it. They communicate with experts to solve a particular problem. The structural definition makes clear that IS are **socio-technical systems**, i.e., systems consisting of humans, behaviour rules, and conceptual and technical artefacts.
An information system can be defined technically as a set of interrelated components that collect (**or retrieve**), process, store, and distribute information to support decision making and control in an organization. In addition to supporting decision making, coordination, and control, information systems may also help managers and workers analyse problems, visualize complex subjects, and create new product.

There are four conceptualizations of the definitions of Information systems according to the literature review by Boell and Cecez-Kecmanovic (2015). They include:

1. The Technical view
2. The Social view
3. The Socio-technical view and
4. The Process views

Technical view

The technical view defines **Information Systems**as systems that utilize computer hardware and software; manual procedures; models for analysis, planning, control and decision-making; and a database. The emphasis is on information technology (IT) embedded in organisations. Definitions falling under the technology view stress the importance of IT in an organizational context or the software used for the processing, storage and distribution of data and information.

Definitions associated with this view does not generally deny the importance of other aspects regarding IS, however, they emphasize the importance of technology, especially IT, in the form of hardware, networks and software over other aspects. In this sense, organizations are seen as a context for IT.

Social view

The social view defines **information systems** as a social system, which has information technology embedded in it. The extent to which information technology plays a part is increasing rapidly. But this does not prevent the overall **information system** from being a social system, and it is not possible to design a robust, effective information system, incorporating significant amounts of the technology without treating it as a **social system**.

Definitions taking a social view emphasize the importance of the social nature of IS. Frequently, they also recognize the importance of technology but they generally consider technology to be subordinate to social aspects. In contrast to the technology view, which locates agency in technology, the social view puts agency on humans and social systems. Therefore, of key importance are the social institutions and organizations that enable and constrain human agency

and the ways in which human actors create, share and interpret information and attribute meanings to IS.

Socio-technical view

Under this view, the **information systems** field examines more than just the technological system, or just the **social system**, or even the two side by side; in addition, it investigates the phenomena that emerge when the two interact. The definitions which fall under the sociotechnical view describe IS in terms of both social and technical aspects that are in continuous interaction. Importantly IS are not only seen as consisting of technological as well as social components, but as phenomena that emerge when they interact. IS are neither technically determined nor socially determined. Instead technology and social systems interact with each other in a way that makes the resulting IS more than the sum of its parts.

In-text Question

What are the four conceptualizations of the definition of information systems according to the literature review by Boell and Cecez-Kecmanovic (2015)?

Answer

1. Technical view,
2. Social view,
3. Socio-technical view and
4. Process view

2.2 Components of an Information System

Every system requires input data. For example, your computer receives data when you press a key or click a menu command. In an information system, data consists of basic facts that are the **system's** raw material. Information is data that has been transformed into output that is valuable to users. An information system has five key components:

2.2.1 Hardware

Hardware consists of everything in the physical layer of the **information system**. For example, hardware can include servers, workstations, networks, telecommunications equipment, fibre-optic cables, mobile devices, scanners, digital capture devices, and other technology-based infrastructure.

Hardware Resources include:

Machines: as computers and other equipment along with all data media, objects on which data is recorded and saved.

1. Computer systems: consist of variety of interconnected peripheral devices. Examples are microcomputer systems, midrange computer systems, and large **computer systems**.

2.2.2 Software Resources

We have previously discussed that,Software refers to the programs that control the hardware and produce the desired information or results. Software consists of system software and application software. **System software** manages the hardware components, which can include a single workstation or a global network with many thousands of clients.

You may have come across this information that;Software Resources includes all sets of information processing instructions. This generic concept of software includes not only the programs, which direct and control computers but also the sets of information processing (**procedures**). Software Resources includes:

1. System software, such as an operating system
2. Application software, which are programs that direct processing for a particular use of computers by end users.
3. Procedures, which are operating instructions for the people, who will use an **information system**. Examples are instructions for filling out a paper form or using a particular software package.

This may not be new to you, in today's interconnected business world, network software is vitally important. Application software consists of programs that support day-to-day business functions and provide users with the information they require. Application software can serve one user or thousands of people throughout an organisation. Examples of company-wide applications, called enterprise applications, include order processing systems, payroll systems, and company communications networks. On a smaller scale, individual users (you) increase their productivity with tools such as spreadsheets, word processors, and database **management systems**. Application software includes horizontal and **vertical systems**. A horizontal system is a system, such as an inventory or payroll application, that can be adapted for use in many different types of companies. A vertical system is designed to meet the unique requirements of a specific business or industry, such as a Web-based retailer, a medical practice, or a video chain.

2.2.3 Data

We have stated earlier that,data is the raw material that an information system transforms into useful information. An **information system** can store data in various locations, called tables. By linking the tables, the system can extract specific information. Data resources include data (which is raw material of information systems) and database.

We have come to understand that, data can take many forms, including traditional alphanumeric data, composed of numbers and alphabetical and other characters that describe business transactions and other events and entities. Text data, consisting of sentences and paragraphs used in written communications; image data, such as graphic shapes and figures; and audio data, the human voice and other sounds, are also important forms of data.

Data resources must meet the following criteria:

1. Comprehensiveness: means that all the data about the subject are actually present in the database.
2. Non-redundancy: means that each individual piece of data exists only once in the database.
3. Appropriate structure: means that the data are stored in such a way as to minimize the cost of expected processing and storage.

The data resources of IS are typically organized into:

1. Processed and organized Data -Databases.
2. Knowledge in a variety of forms such as facts, rules, and case examples about successful business practices.

In-text Question
Define Software Resources

Answer
Software refers to the programs that control the hardware and produce the desired information or results. Software consists of system software and application software.

2.2.4 Network

Telecommunications networks like the Internet, intranets, and extranets have become essential to the successful operations of all types of organisations and their computer-based information systems. Telecommunications networks consist of computers, communications processors, and other devices interconnected by communications media and controlled by communications software. The concept of Network Resources emphasizes that communications networks are a fundamental resource component of all **information systems**.

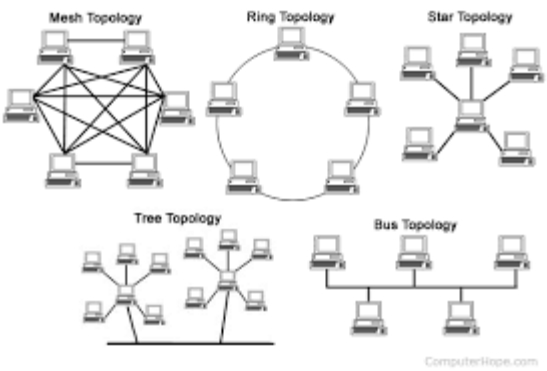


Fig 1.3.2: Network

Source: computerhope.com

Network resources include:

1. Communications media: such as twisted pair wire, coaxial cable, fibre-optic cable, **microwave systems**, and communication satellite systems.
2. Network support: This generic category includes all of the people, hardware, software, and data resources that directly support the operation and use of a communications network. Examples include communications control software such as **network operating systems** and Internet packages.

2.2.5 People (Human Resources)

You should note that, people who have an interest in an **information system** are called stakeholders. Stakeholders include the management group responsible for the system, the users (**sometimes called end users**) inside and outside the company who will interact with the **system**, and IT staff members, such as systems analysts, programmers, and network administrators who develop and support the system.

Each stakeholder group has a vital interest in the information system, but most experienced IT professionals agree that the success or failure of a system usually depends on whether it meets the needs of its users (you). For that reason, it is essential to understand user requirements and expectations throughout the development process.

These users include:

1. End users: (**also called users or clients**) are people who use an information system or the information it produces. They can be accountants, salespersons, engineers, clerks, customers, or managers. Most of us are information system end users.
2. IS Specialists: people who actually develop and operate information systems. They include **systems analysts**, programmers, testers, computer operators, and other managerial, technical, and clerical IS personnel.

Briefly, systems analysts design information systems based on the information requirements of end uses, programmers prepare computer programs based on the specifications of **systems analysts**, and computer operators operate large computer systems.

In-text Question
Which of the components of system has to do with computer operating system?

Answer
The software component

3.0Conclusion/Summary

This study session intimated students with the various definitions and components of an **information system**

4.0Self-Assessment Questions

1. Why do you think the social view conceptualization of the definition of information system is not enough to capture the essence of information systems?
2. Compare the components of information systems given here with that of any system f your choice?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=pOzSlT2Chi0&t=57s Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2xdQdip and critique it in the discussion forum

6.0 References/Further Readings

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STUDY SESSION 4

Types of Information System

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1- Types of Information System

2.1.1- Transaction Processing Systems

2.1.2- Management Information systems

2.1.3- Decision Support Systems

3.0Study Session Summary and Conclusion

4.0Self-Assessment Questions

5.0Additional Activities (Videos, Animations & Out of Class activities)

6.0References/Further Readings

Introduction:

You are welcome to study session 4, in this session you will be exposed to the types of **information systems**.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

A. Identify the different types of Information systems.

2.0 Main Content

2.1 Types of Information System

O'Brien &Marakas (2007), We have previously discussed that, the applications of **information systems** that are implemented in today's business world can be classified in several different ways. For example, several types of information systems can be classified as either operations (**Support of business operation**)

or (**Support of managerial decision making**). Support of business operation such as transaction processing systems, process control systems and Enterprise collaboration systems (office automation system). Support of managerial decision making such as management information system, decision support system and executive information systems.Rogers et al (2005) there are several categories of information system such as Data Processing Systems, Management **Information Systems**, Decision Support Systems and Executive Information System.

2.1.1 Transaction Processing Systems

We have previously discussed that,every organisation has many ***transactionprocessing systems*** *(TPS)*, which capture and processthe detailed data necessary to update records about the fundamental business operations of the organisation. These systems include order entry, inventory control, payroll, accounts payable, accounts receivable, and the general ledger, to name just a few. The input to these **systems** includes basic business transactions, such as customer orders, purchase orders, receipts, time cards, invoices, and customer payments.

The processing activities include data collection, data editing, data correction, data manipulation, data storage, and document. Transaction processing systems (TPS) are the basic **business systems** that serve the operational level of the organisation. A transaction **processing system** is a computerised system that performs and records the daily routine transactions necessary to the conduct of the business. At the lowest level of the organisational hierarchy we find the transaction **processing systems** that support the day-to-day activities of the business.

Transaction processing **systems** support routine operations associated with customer ordering and billing, employee payroll, purchasing, and accounts payable. The amount of support for decision making that a TPS directly provides managers and workers is low. TPSs work with a large amount of input and output data and use this data to update the official records of the company about such things as orders, sales, customers, and so on. As systems move from transaction processing to management information/decision support and special-purpose information systems, they involve less routine, more decision support, less input and output, and more sophisticated and complex processing and analysis. These higher-level systems require the basic business transaction data captured by the TPS.

Fig1.4.1: Transaction Processing Systems

Source: slideshre.net

2.1.2 Management Information Systems

It is good for you to note that,decision making is one of essential management tasks. Effective decision making is *informed* decision making. Managers get informed via **information systems**, oral communication, and possibly in other ways. Management **information systems** are a kind of computer information systems that could collect and process information from different sources in institute decision making in level of management. Management information systems Provide information in the form of pre-specified reports and displays to support business decision making. The next level in the organisational hierarchy is occupied by low level managers and supervisors.

From the above, you will realise,his level contains **computer systems** that are intended to assist operational management in monitoring and controlling the

transaction processing activities that occur at clerical level. Management **information systems** (MIS) use the data collected by the TPS to provide supervisors with the necessary control reports. Hasan (2013) management information system is type of **information systems** that take internal data from the system and summarised it to meaningful and useful forms as management reports to use it to support management activities and decision making.

A management **information system** (MIS) is an integrated collection of people, procedures,databases, and devices that provides managers and decision makers with information to helpachieve organisational goals. MISs can often give companies and other organizations a competitiveadvantage by providing the right information to the right people in the right format and at the right time.

The primary purpose of an MIS is to help an organization achieve its goals by providing managers with insight into the regular operations of the organization so that they can control, organize, and plan more effectively. One important role of the MIS is to provide the right information to the right person in the right format at the right time. In short, an MIS provides managers with information, typically in reports, that supports effective decision making and provides feedback on daily operations. Examples of MISs that are in use in organizations include:

A. Accounting MISs

As we have said, in some cases, accounting works closely with financial management. An **accounting MIS** performs a number of important activities, providing aggregate information on accounts payable, accounts receivable, payroll, and many other applications. The organisation's enterprise resource planning and transaction processing **system** captures accounting data, which is also used by most other functional **information systems**.

Some smaller companies hire outside accounting firms to assist them with their accounting functions. These outside companies produce reports for the firm using raw accounting data. In addition, many excellent integrated accounting programs are available for personal computers in small companies. Depending on the needs of the small organisation and its staff's computer experience, using these computerised **accounting systems** can be a very cost-effective approach to **managing information**.

B. Geographic Information Systems

Let us discuss geographical **information system**.

Increasingly, managers want to see data presented in graphical form. A **geographic information system (GIS)** is a computer system capable of assembling, storing, manipulating, and displaying geographically referenced information, that is, data identified according to its location. A GIS enables users to pair maps or map outlines with tabular data to describe aspects of a particular geographic region. Let us look at this example, sales managers might want to plot total sales for each county in the states they serve.

2.1.3 Decision Support Systems

A **Decision Support System** is a computer-based system intended for use by a particular manager or usually a group of managers at any organisational level in making a decision in the process of solving a semi structured decision. Heidarkhani, et al. (2013) Decision Support Systems are a kind of organisational information computerise systems that help manager in decision making that needs modelling, formulation, calculating, comparing, selecting the best option or predict the scenarios".

Let us look at the work of Khanore, et al. (2011), decision-support systems are specifically designed to help management make decisions in situations where there is uncertainty about the possible outcomes of those decisions. According to Shim (2000) a **decision support system** is a computer-based information system that assists managers in making many complex decisions, such as decisions needed to solve poorly defined or semi-structured problems.

A DSS is an organised collection of people, procedures, software, databases, and devices used to help make decisions that solve problems. The focus of a DSS is on decision-making effectiveness when faced with unstructured or semi-structured business problems.

As with a TPS and an MIS, we can say that, DSS should be designed, developed, and used to help an organisation achieve its goals and objectives. **Decision support systems** offer the potential to generate higher profits, lower costs, and better products and services. From above we have discovered that, for example, healthcare organisations use DSSs to improve patient care and reduce costs. Decision support systems, although skewed somewhat toward the top levels of management, are used at all levels. To some extent, we have also discover that today's managers at all levels are faced with less structured, non-routine problems, but the quantity and magnitude of these decisions increase as a manager rises higher in an organization.

Looking at what we just discussed from the last paragraph we have found out that many organisations contain a tangled web of complex rules, procedures, and decisions. We can see that, DSSs are used to bring more structure to these problems to aid the **decision-making process**. In addition, because of the inherent flexibility of **decision support systems**, managers at all levels are able to use DSSs to assist in some relatively routine, programmable decisions in lieu

of more formalized management **information systems**. DSSs are also used in government, law enforcement, and non-profit organisations.

Characteristics of a Decision Support System

Decision support systems have many characteristics that allow them to be effective management support tools. Of course, not all DSSs work the same. The following list shows some important characteristics of a DSS.

A. Provide rapid access to information.
B. Handle large amounts of data from different sources.
C. Provide report and presentation flexibility.
D. Offer both textual and graphical orientation.
E. Support drill-down analysis.
F. Perform complex, sophisticated analysis and comparisons using advanced software packages.
G. Support optimization, satisfying, and heuristic approaches.
H. Perform simulation analysis—the ability of the DSS to duplicate the features of a real system, where probability or uncertainty is involved.

In-text Question
List any two types of information system?

Answer
Management Information system and transaction Processing System

3.0 Conclusion/Summary

This study session provided an in-depth understanding of types of **information systems** like the **management information systems** and the transaction processing system. The session also provided the students with an understanding of decision support system and its important characteristics.

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=HrrfnPVaG9Q and

https://www.youtube.com/watch?v=73Z3-ekxS7Q Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2DRCobH and critique it in the discussion forum

6.0 References/Further Readings

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MODULE 2

Contents:

Study Session 1: System Analysis.

Study Session 2: Types of System Analysis.

Study Session 3: Who is a System Analyst?

Study Session 4: Skills of a System Analyst.

STUDY SESSION 1

System Analysis

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

- 2.1-Definition of System Analysis
- 2.2- Requirement Determination
- 2.3- Requirement Elicitation Techniques

3.0Study Session Summary and Conclusion

4.0Self-Assessment Questions

5.0Additional Activities (Videos, Animations & Out of Class activities)

6.0References/Further Readings

Introduction:

You are welcome to study module 2, in this study session, you will learn and explore the different definitions of **system analysis**. You will also be exposed to **system requirements** and how they are elicited.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Define system analysis

2. Discuss requirement determination and is place in systems analysis and design.
3. Discuss how requirements are elicited from the different stakeholders in systems analysis and design in organisations

2.0 Main Content

2.1 Definitions of System Analysis

System analysis and design refers to the process of examining a business situation with the intent of improving it through better procedures and methods. **System analysis** and design relates to shaping organisations, improving performance and achieving objectives for profitability and growth. From above you can say that, **system analysis** is an inquiry to aid the decision maker to choose a course of action by systematically investigating his proper objectives, comparing quantitatively wherever possible the costs effectiveness and risks associated with the alternative policies or strategies for achieving them.

It is also a structured technique for handling large, complex problems to lead to an efficient allocation of resources to meet well defined goals and objectives.

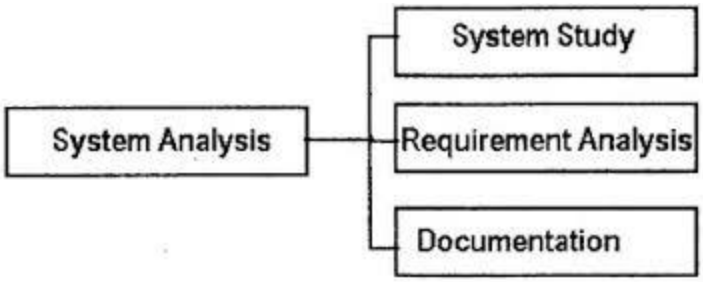

Fig 2.1.1: System Analysis

Source: sciencehq.com

The International Institute of **Applied Systems Analysis** (IIASA) emphasizes that **systems analysis** it is not synonymous with modelling, not about exposing weakness and is not purely scientific or mathematical in nature. According to IIASA "**systems analysis** is a problem-solving process in which many people take part; scientists of relevant disciplines, stakeholders and decision makers.

These are not just problems per se, but problems along with all the attendant factors and concepts they encompass". (Hordijk, 2007).

Quade& Boucher (1968) defined **System Analysis** "as a systemic approach to helping a decision maker choose a course of action by investigating his problem, searching out objectives and alternatives, and comparing them in the light of their consequences, using an appropriate framework - in so far as possible analytic to bring expert judgement and intuition to bear on the problem".

Information systems are created to solve problems. We need to see all sides of a problem to come up with an acceptable solution. Analysis involves studying the system and seeing how they interact with the entities outside as well as inside the system. We then come out with detailed specifications of what the system will accomplish based on the user requirements.System analysis strategy usually includes a study of the current system (**called the *as-is system***) and its problems, and envisioning ways to design a new system (called the *to-**be-system***). This is followed by Requirement determination.

In-text Question
How did Quade& Boucher (1968) define system analysis?

Answer
Quade& Boucher (1968) defined System Analysis "as a systemic approach to helping a decision maker choose a course of action by investigating his problem, searching out objectives and alternatives, and comparing them in the light of their consequences, using an appropriate framework - in so far as possible analytic - to bring expert judgement and intuition to bear on the problem"

2.2 Requirement Determination

You will observe that, the purpose of requirements determination is to turn the very high-level explanation of the business requirements stated in the system request into a more precise list of requirements that can be used as inputs to the rest of analysis (**creating functional, structural, and behavioural models**). This expansion of the requirements ultimately leads to the design of the system.

A requirement is simply a statement of what the system must do or what characteristic it must have. During analysis, requirements are written from the perspective of the businessperson, and they focus on the "**what**" of the system. Because they focus on the needs of the business user, they are usually called business requirements (**and sometimes user requirements**).

Later in design, we observe that,business requirements evolve to become more technical, and they describe how the system will be implemented. Requirements in design are written from the developer's perspective, and they are usually called system requirements. We equally observe that requirements can be either functional or non-functional in nature. You can see that afunctional requirement relates directly to a **process a system** has to perform or information it needs to contain. Let us look at this example, requirements stating that a system must have the ability to search for available inventory or to report actual and budgeted expenses are functional requirements. Functional requirements flow directly into the creation of functional, structural, and behavioural models that represent the functionality of the evolving system. Non-functional requirements refer to behavioural properties that the system must have, such as performance and usability. The ability to access the system using a Web browser is considered a non-functional requirement. Non-functional requirements can influence the rest of analysis (**functional, structural, and behavioural models**) but often do so only indirectly; non-functionalrequirements are used primarily

in design when decisions are made about the database, the user interface, the hardware and software, and the system's underlying physical architecture.

In-text Question
Why are requirements important during systems analysis?

Answer
A requirement is simply a statement of what the system must do or what characteristic it must have. During analysis, requirements are written from the perspective of the businessperson, and they focus on the "what" of the system.

2.4 Requirement Elicitation Techniques

It is important for you to note that, the requirements-elicitation process is used for building political support for the project and establishing trust and rapport between the project team building the **system** and the users who ultimately will choose to use or not use the **system**. Involving someone in the process implies that the project teams view that person as an important resource and value his or her opinions.

Fig 2.1.2: Requirement Elicitation Techniques
Source: iag.biz

All the key stakeholders (**the people who can affect the system or who will be affected by the system**) must be included in the requirements-gathering process. The stakeholders might include managers, employees, staff members, and even some customers and suppliers. If a key person is not involved, that individual might feel slighted, which can cause problems during implementation (e.g., How could they have developed the system without my input?).

The second challenge of requirements elicitation is choosing the way(s) information is collected. There are many techniques for eliciting requirements that vary from asking people questions to watching them work. The following are some of the elicitation techniques you can use:

1. Interviews: An interview is the most commonly used requirements-gathering technique. After all, it is natural. If you need to know something, you usually ask someone. In general, interviews are conducted one-on-one (**one interviewer and one interviewee**), but sometimes, owing to time constraints, several people are interviewed at the same time.
2. Joint Application Development (JAD): JAD is an information-gathering technique that allows the project team, users, and management to work together to identify requirements for the system. IBM developed the JAD technique in the late 1970s, and it is often the most useful method for collecting information from users. JAD is a structured process in which ten to twenty users meet together under the direction of a ***facilitator*** skilled in JAD techniques. The facilitator sets the meeting agenda and guides the discussion but does not join in the discussion as a participant. He or she does not provide ideas or opinions on the topics under discussion so as to remain neutral during the session. The facilitator must be an expert in both group-process techniques and systems analysis and design techniques. One or two ***scribes***assist the facilitator by recording notes, making copies, and so on. The JAD group meets for several hours, several days, or several weeks until all the issues have been discussed and the needed information is collected. Most JAD sessions take place in a specially prepared meeting room, away from the participants' offices so that they are not interrupted. The meeting room is usually arranged in a U-shape so that all participants can easily see each other. At the front of

the room (the open part of the U), are a whiteboard, flip chart, and/or overhead projector for use by the facilitator leading the discussion.

3. Questionnaires: A questionnaire is a set of written questions used to obtain information from individuals. Questionnaires are often used when there is a large number of people from whom information and opinions are needed. In our experience, questionnaires are a common technique with systems intended for use outside the organization (e.g., by customers or vendors) or for systems with business users spread across many geographic locations. Most people automatically think of paper when they think of questionnaires, but today more questionnaires are being distributed in electronic form, either via e-mail or on the Web. Electronic distribution can save a significant amount of money as compared to distributing paper questionnaires.
4. Document Analysis: Project teams often use document analysis to understand the as-is system. Under ideal circumstances, the project team that developed the existing **system** will have produced documentation that was then updated by all subsequent projects. In this case, the project team can start by reviewing the documentation and examining the system itself. Unfortunately, many systems are not well documented because project teams fail to document their projects along the way, and when the projects are over, there is no time to go back and document. Therefore, there might not be much technical documentation about the current **systems available**, or it might not contain updated information about recent system changes.
5. Observation: Observation, the act of watching processes being performed, is a powerful tool for gathering information about the as-is **system** because it enables the analyst to see the reality of a situation, rather than listening to others describe it in interviews or JAD sessions. Several research studies have shown that many managers really do not remember

how they work and how they allocate their time. (**Quick, how many hours did you spend last week on each of your courses?**) Observation is a good way to check the validity of information gathered from indirect sources such as interviews and questionnaires. In many ways, the analyst becomes an anthropologist as he or she walks through the organization and observes the business system as it functions. The goal is to keep a low profile, to not interrupt those working, and to not influence those being observed. Nonetheless, it is important to understand that what analysts observe may not be the normal day-to-day routine because people tend to be extremely careful in their behavior when they are being watched.

In-text Question
List five techniques for eliciting requirements in systems analysis

Answer
Joint Application Development, Interviews, Questionnaire, Document Analysis and Observation

3.0 Conclusion/Summary

This study session exposed you to the different definitions of system analysis. It also intimated students on the importance of requirement determination in **system analysis** and the techniques for eliciting these requirements.

4.0 Self-Assessment Questions

1. Identify any **information system** of your choice any state its requirement from your point of view as a user?
2. Relate how the information systems in your organizations are analysed?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=pOzSlT2Chi0&t=143s Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2WpuRId and critique it in the discussion forum

6.0 References/Further Readings

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STUDY SESSION 2

Methodologies of System Analysis

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1-Methodologies of Systems Analysis

2.1.1-Formulation

2.1.2- Search

2.1.3- Explanation

2.1.4- Evaluation

2.1.5- Interpretation

2.1.6- Implementation

3.0Study Session Summary and Conclusion

4.0Self-Assessment Questions

5.0Additional Activities (Videos, Animations & Out of Class activities)

6.0 References/Further Readings

Introduction:

You are welcome to study session2 in this session, you will be exposed to the **methodologies of system analysis**.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to

1. Discuss the methodologies of system analysis.

2.0 Main Content

2.1 Methodologies of Systems Analysis

We are going to look at the six basic activities in analysis that can serve a framework for a discussion of the **methodology of systems analysis**. These are: Formulate, Search, Explain, Evaluate, Interpret and implement. These activities are discussed below:

2.1.1 Formulation

This activity is basically one of problem definition. A problem should not be accepted as originally stated and a considerable effort should go into understanding and formulating the problem. A proper problem definition should consider the goals and objectives of the **system** as well as its inputs, outputs, variables, constraints, parameters and relationships. Questions that should be asked include: What are the functions that the system you are studying tries to perform? What is deficient in the status quo? What are the relevant factors to the problem (**there very easily could be hundreds of them**)? What are the higher and lower order systems? What resources do you have to study the problem with and to implement solutions with? Can you generalize the problem in a broad sense? All these questions can help to provide insight into the nature of a problem.

One should be very careful not to let statements of alternative solutions become part of a problem statement. For example, the problem statement build a better mousetrap is a poor one since it mentions a solution the mousetrap. A better statement might be to develop an effective rodent control system. Then the mousetrap and other rodent control techniques could be considered as possible solutions to the problem.

2.1.2 Search

Fig 2.2.1: Search

This phase involves such activities as finding facts and data, searching for relationships, and the assessment of relevant factors. Usually information and data will cost money and shouldn't be collected unless one is certain it is needed. It is important to maintain a proper perspective as to how the information will be used to avoid information overload.

The development of alternatives is largely a creative art, but there are certain principles that may help. There should be a very large number of alternatives initially and they should vary widely in scope. By avoiding criticism in their development it is possible to develop a large quantity of alternatives. It is desirable to think in terms of unevaluated alternatives since a later phase of the **analysis cycle** will be used to give explicit rationale for their elimination. The premature elimination of alternatives is undesirable since it may often be done for poor reasons and potentially good alternatives may be lost. Alternatives may involve changes in technology, changes in policies, changes in procedures or changes in constraints. It is desirable to look at the extremes, and try to develop a continuum between them and to look for basic elements of a **system** and the forms they may take and recombine them in new ways.

2.1.3 Explanation

One of the central phases of analysis is to develop theories or models to aid in explanation of **system behaviour**. A model is a small-scale version of the real world, a simplification. It is developed for some specific purpose and its form is based on this purpose, e.g., form follows function. Ideally the results of the

model should closely match what would occur in the real world. Nevertheless, it is only an indicator and the results should be carefully interpreted. Assumptions should be stated explicitly.

Models can be built in two ways, by contraction or by expansion. To build a model by contraction, you first attempt to find all the variables that may be important and systematically reduce the list until a model of reasonable size results. The method of expansion involves the building of initial models under simplifying assumptions. The assumptions are relaxed (**e.g., constants become variables**) and the model is expanded to the desired level of detail.
A model can be a useful tool of communication or an aid to understanding a process as well as a direct tool in the analysis.

2.1.4 Evaluation

Alternatives should be compared on the basis of their costs and their effectiveness. Both have to be considered for meaningful results. Both costs and effectiveness should be defined rather broadly, perhaps to include such things as social costs, and secondary benefits. The quality of the evaluation is enhanced if alternatives are compared in a consistent manner. One would be concerned with getting the greatest effectiveness for a given budget or a fixed level of gain for the least cost.

Evaluation should also include the assessment of sensitivities. This gives clues as to what further data should be collected or how strong the choice is. The feasibility of the alternatives should also be considered. This would include political, economic, social, technical and institutional feasibilities as well as the technical feasibility. The evaluation also should take note of the impact and incidence of costs and benefits and when the costs and benefits occur.

2.1.5 Interpretation

Results of the analysis should be interpreted in light of factors that were not directly considered. Results should not be accepted blindly, but consideration should be given to non-quantifiable, omitted factors, uncertainties, and contingencies. These things might have a significant effect on eventual success.

Fig 2.2.2: Interpretation
Source: communicationdubai.com

Uncertainty should be treated explicitly to determine its effect upon the best choice. Likewise, one should consider how his recommended solution will act under unusual circumstances which are possible but not probable, i.e., contingencies.

Confidence in a choice may be increased by a break-even analysis, i.e., how far can you be off before the recommendations change.

2.1.6 Implementation

The final phase of the **analysis process** is one of implementation. In many ways this phase is inherent in the other activities of the analysis. Solutions that cannot be implemented are of very little value. Implementation is aided by careful planning and an understanding of the mechanisms for implementation. In this phase as well as all other phases of an analysis careful planning is vitally important. This phase also involves the modification and adaptation of results to new conditions and circumstances. It also involves updating of data, re-evaluation of procedures and techniques, expansion of results to other uses and applications and other activities.

In-text Question

List the six basic approaches to system analysis?

Answer
Formulate, Search, Explain, Evaluate, Interpret and implement

3.0 Conclusion/Summary

This study session provided students with the understanding of methodologies of **system analysis**.

4.0 Self-Assessment Questions

1. Why should considerable effort be expended on the formulation of a problem during system analysis

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=pOzSlT2Chi0&t=143s Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2TGWYAs and critique it in the discussion forum

6.0 References/Further Readings

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STUDY SESSION 3

Who is a System Analyst?

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1-Who is a System Analyst?

2.2- Other Analysts involved in Information Systems Analysis and Design

2.2.1- Business Analyst

2.2.2- Infrastructure Analyst

2.2.3- Change Management Analyst

3.0Study Session Summary and Conclusion

4.0Self-Assessment Questions

5.0Additional Activities (Videos, Animations & Out of Class activities)

6.0 References/Further Readings

Introduction:

You are welcome to study session 3, in this session you will learn who a **system analyst** is andyou will also be exposed to other analysts that are involved in **information systems analysis** and design.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Identify who a system analysis is
2. Identify and discuss the responsibilities of other analysts that are involved in information systems analysis and design.

2.0 Main Content

2.1 Who is a System Analyst?

Who is a **system analyst**? You can define a systems analyst as a person who investigates,analyses designs, develops, installs, evaluates, and maintains a company's **information systems**. To perform those tasks, a systems analyst constantly interacts with users and managers within and outside the company. On large projects, the analyst works as a member of an IT department team; on smaller assignments, he or she might work alone.

Fig 2.3.1: Who is a System Analyst?
Source: rickiehui.wordpress.com

You will observe that,most companies assign systems analysts to the IT department, but analysts also can report to a specific user area such as marketing, sales, or accounting. You as a member of a functional team, an analyst is better able to understand the needs of that group and how **information systems** support the department's mission. Smaller companies often use consultants to perform systems analysis work on an as-needed basis.

The System analyst is an individual in charge of designing, modifying, or analysing various **systems** to ensure compatibility and user effectiveness. **System analysts** may work independently, but are often part of a larger information technology unit that keeps technical aspects of a company running smoothly. **System analysts** do not focus on hard-coding, but more on evaluating the code and making suggestions. **Systems analysts** are **information systems** professionals who specialize in analysing and designing information systems. Systems analysts are responsible for the complete life-cycle of a

new/modified IT **system**, from analysing existing arrangements to implementing **systems** and providing training. Employers often require candidates to meet high standards in programming aptitude tests. You find out that a systems analyst combines knowledge of information systems and business to develop computer systems that will meet business needs. The job is among the most advanced computer careers because of your need to be able to think strategically and to interact with professionals with a variety of skill sets.

In-text Question
Who is a system analyst?

Answer
The System analyst is an individual in charge of designing, modifying, or analyzing various systems to ensure compatibility and user effectiveness

2.2 Other Analysts involved in Information System Analysis and Design

The following are other analysts that are involved in information **systems analysis and design**:

2.2.1 Business Analyst

A **business analyst**focuses on the business issues surrounding the **system**. These issues include identifying the business value that the system will create, developing ideas and suggestions for how the business processes can be improved, and designing the new processes and policies in conjunction with the **systems analyst**. This individual likely has business experience and some type of professional training.

He or she represents the interests of the project sponsor and the ultimate users of the system. A business **analyst** assists in the planning and design phases but is most active in the **analysis phase**.

2.2.2 Infrastructure Analyst

As stated earlier,we say an infrastructure analyst focuses on the

technical issues surrounding how the **system** will interact with the organisation's technical infrastructure (**e.g., hardware, software, networks, and databases**). An infrastructure analyst's tasks include ensuring that the new information system conforms to organisational standards and identifying infrastructure changes needed to support the **system**.

You find out that this individual probably has significant training and experience in networking, database administration, and various hardware and software products. He or she represents the interests of the organisation and IS group that will ultimately have to operate and support the new system once it has been installed. An infrastructure analyst works throughout the project but perhaps less so during planning and analysis phases.

2.2.3 Change Management Analyst

A change management analyst focuses on the people and management issues surrounding the **system installation**. You should know that the roles of this person include ensuring that the adequate documentation and support are available to users, providing user training on the **new system**, and developing strategies to overcome resistance to change. This individual should have significant training and experience in organisational behaviour in general and change management in particular. He or she represents the interests of the project sponsor and users for whom the **system** is being designed. A change management analyst works most actively during the implementation phase but begins laying the groundwork for change during the analysis and design phases.

In-text Question
What are the roles of a change management analyst?

Answer

The roles of the Change management analyst include ensuring that the adequate documentation and support are available to users, providing user training on the new system, and developing strategies to overcome resistance to change.

3.0 Conclusion/Summary

This study session provided students with the knowledge of the skills of a **system analyst** and other analysts that organizations should have.

4.0 Self-Assessment Questions

1. Why should systems analysts be attached to IT departments of organizations?
2. How does the work of an infrastructure analyst differ from that of the system analyst?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=pOzSlT2Chi0&t=143s Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2WpuRId and critique it in the discussion forum

6.0 References/Further Readings

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STUDY SESSION 4

Skills of System Analysts

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1-Skills of a System Analyst

2.1.1- Technical Skills

2.1.2- Communication Skills

2.1.3- Business Skills

2.1.4- Critical Thinking Skills

2.1.5- Education/Certification

3.0Study Session Summary and Conclusion

4.0Self-Assessment Questions

5.0Additional Activities (Videos, Animations & Out of Class activities)

6.0 References/Further Readings

Introduction:

You are welcome to Study Session4, in this session you will be exposed to the **skills set of a system analyst**. This includes the technical skills, communication skills, business skills, critical thinking skills and education/certification.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to

1. Discuss the skills of a system analyst

1.0 Main Content

2.0 Skills of a System Analyst

Let us discuss the skills of a system analyst

The **systems analyst's** job overlaps business and technical issues. Analysts help translate business requirements into IT projects. From the above, you will realise, that when assigned to a **systems development team**, an analyst might help document business profiles, review business processes, select hardware and software packages, design information systems, train users, and plan e-commerce Web sites. A systems analyst plans projects, develops schedules, and estimates costs. To keep managers and users informed, the analyst conducts meetings, delivers presentations, and writes memos, reports, and documentation.

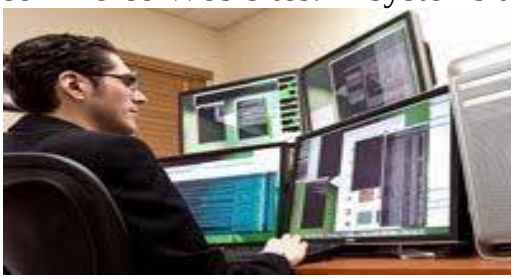

Fig 2.4.1: Skills of a System Analyst

You observe that a successful **systems analyst** needs technical knowledge, oral and written communication skills, an understanding of business operations, and critical thinking skills. Educational requirements vary widely depending on the company and the position. In a rapidly changing IT marketplace, a systems analyst must manage his or her own career and have a plan for professional development.

The following are the skill sets of a System Analyst:

2.1.1 Technical Skills

State-of-the-art knowledge is extremely important in a rapidly changing business and technical environment. The Internet offers numerous opportunities to update technical knowledge and skills. **Analysts** also maintain their skills by attending training courses, both on-site and online. Your networking with colleagues is another way to keep up you with new developments, and

membership in professional associations also is important. Analysts must have the technical skills to understand the organisation's existing technical environment, the new system's technology foundation, and the way in which both can be fit into an integrated technical solution.

2.1.2 Communication Skills

As an analyst,you need to communicate effectively, one-on-one with users and business managers (**who often have little experience with technology**) and with programmers (**who often have more technical expertise than the analyst does**). They must be able to give presentations to large and small groups and to write reports. You as a**systems analyst** needs strong oral and written communication skills, and the ability to interact with people at all levels, from operational staff to senior executives. Often, the analyst must work with people outside the company, such as software and hardware vendors, customers, and government officials. **Analysts** often coordinate IT project teams, where they use communication skills to guide and motivate team members.

2.1.3 Business Skills

It is good to note that, a systems analyst works closely with managers, supervisors, and operational employees. To be effective, you (**he or she**) must understand business operations and processes, communicate clearly, and translate business needs into requirements that can be understood by programmers and systems developers. A successful **analyst** is business-oriented, curious, comfortable with financial tools, and able to see the big picture.

Fig 2.4.2: Business Skills

Source: meaningfulwoman.com

Business skills are required to understand how IT can be applied to business situations and to ensure that the IT delivers real business value. **Analysts** are continuous problem solvers at both the project and the organizational level, and they put their **analytical skills** to the test regularly.

2.1.4 Critical Thinking Skills

What do you understand by critical thinking skills?

Although no standard definition exists, most educators agree that critical thinking skills include the ability to compare, classify, evaluate, recognize patterns, analyse cause-and-effect, and apply logic. Critical thinkers often use a ***what-if***approach, and they have the ability to evaluate their own thinking and reasoning. Critical thinking skills are valuable in the IT industry, where employers seek job candidates who can demonstrate these skills and bring them to the workplace.

2.1.5 Education/Certification

Companies typically require **systems analysts** to have a college degree in information systems, computer science, or business, and some IT experience usually is required. For higher-level positions, many companies require an advanced degree. Sometimes, educational requirements can be waived if a candidate has significant experience, skills, or professional certifications.

In addition to education, the system analyst should possess hardware and software companies offer certification for IT professionals. **Certification** verifies that an individual demonstrated a certain level of knowledge and skill on a standardized test. Certification is an excellent way for IT professionals to learn new skills and gain recognition for their efforts. Although certification

does not guarantee competence or ability, many companies regard certification as an important credential for hiring or promotion.

In-text Question
List the skills that a system analyst should possess?

Answer
Technical skills, communication skills, business skills, critical thinking skills and education/certification

3.0 Conclusion/Summary

This study session provided students with the skills of a **system analyst**.

4.0 Self-Assessment Questions

1. Briefly discuss how each of the skills of the system analyst will profit the organization during **systems analysis and design**.

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=UuCuC1ZOnrs Watch the video & summarise in 1 paragraph

b. View the animation on add/site https://bit.ly/2N26QDf and critique it in the discussion forum

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MODULE 3

Information System Design and Development Lifecycle

Contents:

Study Session 1: Definition of System Design

Study Session 2: System Design tools

Study Session 3: System Development Lifecycle (SDLC)

Study Session 4: System Development Models

STUDY SESSION 1

Definition of System Design

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

- 2.1 Definition of System Design
  - 2.1.1 Types of System Design
- 2.2 System Design Methodologies
  - 2.2.1 Top Down
  - 2.2.2 Bottom up

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos, Animations & Out of Class activities)

6.0 References/Further Readings

Introduction:

You are welcome to study session 1, this will be our last study module of this course I hope you have gained so much from the two previous modules where you were exposed to various topics.

This study session will intimateyou to the definition of **system design** and the types of **system design**. This study session will also expose you to system **design methodologies**.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the system design phase
2. Identify the types of system designs
3. Discuss system design methodologies.

2.0 Main Content

2.1 Definition of System Design

In a traditional SDLC environment, **systems design** usually starts afterthe systems analysis phase. Using the system requirements specification (SRS) as a blueprint, developers transform the **logical design** into a working model that could be tested, reviewed by users, and implemented. It is at the **system design** stage that the SRS document is converted into a format that can be implemented and decides how the **system** will operate.

Fig 3.1.1: Definition of System Design

System design is the process of defining the elements of a **system** such as the architecture, modules and components, the different interfaces of those components and the data that goes through that **system**. It is meant to satisfy specific needs and requirements of a business or organisation through the engineering of a coherent and well-running system.

System design is the phase that bridges the gap between problem domain and the existing **system** in a manageable way. This phase focuses on the solution domain, i.e. *"**how to implement**?"*.In this phase, the complex activity of **system development** is divided into several smaller sub-activities, which coordinate with each other to achieve the main objective of **system development**.

The goal of systems design is to build a system that satisfies business requirements. A successful **system** must be effective, reliable, and maintainable:

1. A system is effective if it supports business requirements and meets user needs.
2. A system is reliable if it handles input errors, processing errors, hardware failures, or human mistakes. A good design will anticipate errors, detect them as early as possible, make it easy to correct them, and prevent them from damaging the system itself.
3. A system is maintainable if it is flexible, scalable, and easily modified. Changes might be needed to correct problems, adapt to user requirements, or take advantage of new technology.

The ***design phase***decides *how* the system will operate, in terms of the hardware, software, and network infrastructure; the user interface, forms, and reports; and the specific programs, databases, and fi les that will be needed. Although most of the strategic decisions about the **system** were made in the development of the system concept during the analysis phase, the steps in the design phase determine exactly how the system will operate. Thedesign phase has four steps:

1. The ***design strategy***is first developed. It clarifies whether the system will be developed by the company's own programmers, whether the system will be outsourced to another firm (**usually a consulting firm**), or whether the company will buy an existing software package.

2. This leads to the development of the basic ***architecture design***for the system, which describes the hardware, software, and network infrastructure to be used.

In most cases, the system will add or change the infrastructure that already exists in theorganization. The ***interface design***specifies how the users will move through the system (e.g., **navigation methods such as menus and on-screen buttons**) and the forms and reports that the system will use.

3. The ***database and file specifications***are developed. These define exactly what data will be stored and where they will be stored.

4. The analyst team develops the ***program design***, which defines the programs that need to be written and exactly what each program will do.

This collection of deliverables (**architecture design, interface design, database and file specifications, and program design**) is the ***system specification***that is handed to the programming team for implementation. At the end of the design phase, the feasibility analysis and project plan are re-examined and revised, and another decision is made by the project sponsor and approval committee about whether to terminate the project or continue.

The systems design approach first appeared right before World War II, when engineers were trying to solve complex control and communications problems. They needed to be able to standardize their work into a formal discipline with proper methods, especially for new fields like information theory, operations research and computer science in general.

2.1.1 Types of System Design

The following are the types of system design:

Logical Design

It is important for you to note that, logical design pertains to an abstract representation of the data flow, inputs, and outputs of the system. It describes the inputs (**sources**), outputs (**destinations**), databases (**data stores**), procedures (**data flows**) all in a format that meets the user(you)requirements. Whilepreparing the logical **design of a system**, the **system analyst** specifies the

user needs at level of detail that virtually determines the information flow into and out of the system and the required data sources. A logical design defines *what* must take place, not *how* it will be accomplished. Logical designs do not address the actual methods of implementation.

Physical Design

Physical design relates to the actual input and output processes of the system. It focuses on how data is entered into a system, verified, processed, and displayed as output. It produces the **working system** by defining the design specification that specifies exactly what the candidate **system** does. It is concerned with user interface design, **process design**, and **data design**.You can see that a physical design is like a set of blueprints for the actual construction of a building. Typically, a physical design describes the actual processes of entering, verifying, and storing data; the physical layout of data files and sorting procedures, the format of reports.

It consists of the following steps −

1. Specifying the input/output media, designing the database, and specifying backup procedures.
2. Planning system implementation.
3. Devising a test and implementation plan, and specifying any new hardware and software.
4. Updating costs, benefits, conversion dates, and system constraints.

Architectural Design

It is also known as high level design that focuses on the **design of system** architecture. It describes the structure and behaviour of the **system**. It defines the structure and relationship between various modules of system development process. Architectural designcentres on how the **information system** components will be distributed across multiple computers and what hardware, operating system software, and application software will be used on each computer (e.g., **Windows or Linux operating system software**). Designing the

system architecture can be quite difficult; therefore, many organizations use the skills of experienced, expert system architects (**consultants or employees**) who specialize in the task.

The non-functional requirements developed early in the analysis phase play a key role in architecture design. These requirements are re-examined and refined into more detailed requirements that influence the system's architecture.

The major ***architectural components***of any system are the software and the hardware. The major software components of the system being developed have to be identified and then allocated to the various hardware components on which the system will operate. Each of these components can be combined in a variety of different ways.

In-text Question
What are the types of system design

Answer
Logical, Physical and Architectural Designs

2.2 System Design Methodologies

Systems design implies a systematic approach to the design of a system. It may take a bottom-up, top-down approach, but either way the process is systematic wherein it considers all related variables of the system that needs to be created from the architecture, to the required hardware and software, right down to the data and how it travels and transforms throughout its travel through the system. **Systems design** then overlaps with **systems analysis**, systems engineering and systems architecture. The following are methodologies of **system design**:

1. **Top down design**

A top-down design is the decomposition of a **system** into smaller parts in order to comprehend its compositional **sub-systems**. In top-down design, a system's overview is designed, specifying, yet not detailing any first-level subsystems. Then, every subsystem is refined in greater detail, for example, sometimes dividing into many different levels of subsystem, so that the whole specification is decomposed to basic elements.

As soon as these base elements are identified, it is easier to build these elements as computer modules. Once the modules are built, it is effortless to put them together, building the whole **system** from these individual elements. A top-down design is also known as a **stepwise design**. A top-down design is generally a plan made in plain, simple English for the program. It is very important to note that a **top-down design** must be independent of any programming language. The top-down design must never incorporate references to library functions or syntactic elements specific to a particular language.

That is the reason why top-down designs are written in plain English. The concept driving a **top-down design** is to break up the task that a program executes into a very few extensive subtasks. The highest level is known as the main module, top level or level 0. At this point, the volume of subtasks must be small. Most programs at this level usually include three to seven subtasks. For small-sized programs, the volume of subtasks must be at the low end of the mentioned range.

Division of tasks into subtasks essentially splits the problem into various smaller programs, which helps developers to easily code these simpler parts. It is usually possible that many of these subtasks are so basic that they can immediately determine how to write the code to complete that part.

However, this is generally not the case at the top level. If a subtask takes more than a very few **lines of code**, it is recommended to repeat the subdivision process. Typically, for every subtask at top level, a new module is started at level 1. These subtasks are then considered individually, and yet again divided into various other subtasks. This subdivision and new level creation processes should be continued until the coders can implement every portion of the problem using computer code.

2. **Bottom up design**

A **bottom-up** approach is the piecing together of **systems** to give rise to more complex systems, thus making the original systems sub-systems of the emergent system. Bottom-up processing is a type of information processing based on incoming data from the environment to form a perception. From a Cognitive Psychology perspective, information enters the eyes in one direction (**sensory input, or the "bottom"**), and is then turned into an image by the brain that can be interpreted and recognized as a perception (**output that is "built up" from processing to final cognition**). In a bottom-up approach, the individual base elements of the system are first specified in great detail. These elements are then linked together to form larger subsystems, which then in turn are linked, sometimes in many levels, until a complete top-level system is formed. This strategy often resembles a "**seed**" model, whereby the beginnings are small but eventually grow in complexity and completeness.

Fig 3.1.2: Definition of System Design

However, "**organic strategies**" may result in a tangle of elements and subsystems, developed in isolation and subject to local optimization as opposed to meeting a global purpose.

If you know what Legos are, you know what bottom-up **design** is. It consists of defining and coding the very basic, definite parts of the **system** to be designed, then linking these parts together to form the whole.

Why Bottom-Up?

1. This approach allows teams to code functioning sub-systems quickly
2. Testing can be done early and often, as first-level systems are defined first
3. It encourages and leads to reusable code
4. Pre-existing code is simpler to incorporate and test

In-text Question
What are the methodologies of system design?

Answer
Top down and Bottom up methodologies

3.0 Conclusion/Summary

This study session provided the students with the definition of **system design** and the methodologies of **system design**.

4.0 Self-Assessment Questions

1. Briefly discuss the major difference between the top down methodology and the bottom up methodology?

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=h7MhEB2ye-g Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2V7QAmQ and critique it in the discussion forum

6.0References/Further Readings

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O'Brien, J. A; &Marakas, George. M. 2007. *Introduction to Information Systems*.

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STUDY SESSION 2

System DesignTools

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1- System Design Tools

2.1.1- Screen Layout Charts

2.1.2- Flowcharts

2.1.3- Decision Tables

2.1.4- Grid Charts

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos, Animations & Out of Class activities)

6.0 References/Further Readings

Introduction:

You are welcome to study session 2, in this session, you will learn **systemdesign tools** to include screen layout charts, flowcharts, **decision tables** and the grid charts.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss the system design tools.

2.0 Main Content

2.1 System Design Tools

This will sound interestingly to know that, a number of tools and techniques are available to computer programmers and **systems analysts** to assist in developing computer programs and applications, in this session, you found out that some of the more traditional tools and techniques are covered. They include print and layout charts, screen layout charts, flowcharts, decision tables, and grid charts.

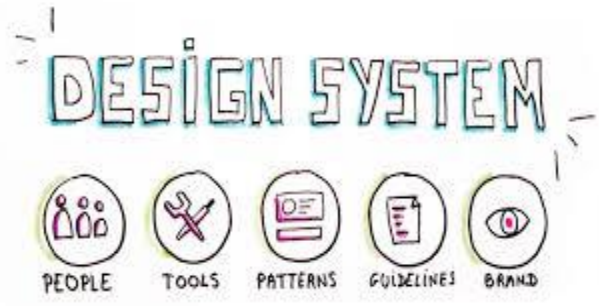

Fig 3.2.1.: System Design Tools

Source: uxdesign.cc

At this stage of its development, you should understand thatthe system is coded, tested, and debugged in a process called programming. Programming is writing instructions for computer execution and testing the written code to ensure that it performs according to specifications. The objective of programming is thus to produce reliable software based on appropriate design specifications. The following are common **system design tools**:

2.1.1 Screen Layout Charts

A number of forms have been developed to make it easier to specify in detail the input and output from a computer application. A print chart is used to describe the format of an output report from a printer. A card layout chart is used to describe how input data is placed on one or more computer cards. Different companies and data processing departments might use slightly different forms.

Screen layout charts can take on a number of different forms, depending upon the detail that is to be displayed on the chart. General screen layout charts reveal an overview of what is to be displayed on the screen. A detailed screen layout

chart reveals in detailed fashion what **information** is to be displayed on the display terminal. Screen layout charts can be done by hand or by using pre-printed forms.

The purpose of a screen layout chart is to reveal what information and data is to be displayed on the screen during processing. The screen layout chart reveals both data to be entered into the computer as well as the information and data to be displayed on the **terminal device**. For some applications, there may be 100 or more different screens that can appear on the **display terminal**. To perform an adequate job during **design and system** documentation, screen layout charts should be developed for every possible screen that can appear on the display terminals.

2.1.2 Flowcharts

This will sound interestingly to note that, just like a road map, flowcharts are used to reveal how to go from a starting point to the final destination and can be used to **display any amount of detail**. In developing application software, a general chart to reveal the overall purpose and structure of the application is needed. This is usually called the **system flowchart** or application flowchart. More detailed flowcharts are also needed that reveal how each program is to be developed. This type of flowchart is called a program flowchart.

Both **system and program** flowcharts are drawn using a special flowcharting template, which contains a set of symbols needed in developing flowcharts. System flowcharts are some symbols are used for both system and program flowcharts, while other symbols are fairly specialized.

Name	Symbol	Description
Process		Process or action step
Flow line		Direction of process flow
Start/ terminator		Start or end point of process flow
Decision		Represents a decision making point
Connector		Inspection point
Inventory		Raw material storage
Inventory		Finished goods storage
Preparation		Initial setup and other preparation steps before start of process flow
Alternate process		Shows a flow which is an alternative to normal flow
Flow line(dashed)		Alternate flow direction of information flow

Fig 3.2.2: Flowcharting Symbols

These system flowchart symbols are used to reveal, in a general fashion, the relationships among the input data, the programs, and the desired output. For example, most realistic payroll applications require multiple computer programs interacting with multiple **data files**.

2.1.3 Decision Tables

A decision table is a logical structure that shows every combination of conditions and outcomes. Analysts often use decision tables to describe a process and ensure that they have considered all possible situations. You can create decision tables using Microsoft PowerPoint, Word, or Excel.

A decision table can be used as an alternative or as an addition to flowcharts. The objective of both decision tables and flowcharts is the same. They reveal to

you how the input data is to be manipulated to obtain the desired output. Just as the name implies, decision tables reveal what **decisions or actions** the computer is to take as a result of the input data. Interestingly, when the computer has to make a large number of decisions or if there are a large number of different branches within a program, decision tables are particularly useful. In these cases, decision tables are preferred to flowcharts. In general, a decision table displays the different conditions that could exist and the different actions that the computer should take as a result of these conditions.

2.1.4 Grid Charts

Application programs are usually complex and interrelated in business. The same data may be used for several programs, and the output from one program may be the input to another. A **grid chart** is an excellent method to reveal the relationships between **application programs**. What data is used for several application programs? What data is used to produce different reports and documents? For which applications is the output from one application used as input for another application? These and similar questions can be answered using grid charts.

In-text Question
List four system design tools?

Answer
Screen Layout Charts, Flowcharts, Decision Tables and Grid Charts

3.0 Conclusion/Summary

This study session provided students with the tools for system design.

4.0 Self-Assessment Questions

1. Briefly discuss when it is appropriate to use any of these tools for systems design

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=pOzSlT2Chi0&t=158s Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2IXR39i and critique it in the discussion forum

6.0 References/Further Readings

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STUDY SESSION 3

System Development Life Cycle

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1- Overview of System Development Lifecycle (SDLC)

2.2- Phases of System Development Lifecycle (SDLC)

2.2.1- Planning

2.2.2- Analysis

2.2.3- Design

2.2.4- Implementation

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos, Animations & Out of Class activities)

6.0 References/Further Readings

Introduction:

You are welcome to study session 3, this session will exposeyouto the development of systems. This study session will also expose you to the different phases of **system development lifecycle**.

1.0 Study Session Learning Outcomes

After you have finished studying this session, you should be able to:

1. Discuss system development lifecycle
2. Discuss the different phases of system development.

2.0Main Content

2.1 Overview of System Development Life Cycle

You may own one or building your first house, developing an information system is similar to building a house. First, building a house(**or the information system**) starts with a basic idea. Then the idea is transformed into a simpleplan in the form of a drawing by the architect. This drawing is then refined jointly with the client until the client becomes satisfied. Third, a model is designed that presents detailed information about the house (e.g., **the type of water faucets or where the telephone jacks will be placed**). Finally, the house is built following the models, often with some changes directed by the client as the house is erected.

Fig 3.3.1: System Development Lifecycle
Source: err.wikipedia.org

We have previously discussed that,the SDLC has a similar set of four fundamental ***phases***: planning, analysis, design, and implementation. Different projects might emphasize different parts of the SDLC or approach theSDLC phases in different ways, but all projects have elements of these four phases. Each *phase* is itself composed of a series of *steps*, which rely upon *techniques* that produce ***deliverables***(**specific documents and files that provide understanding about the project**).

System Development Life Cycle (SDLC) is a conceptual model which includes policies and procedures for developing or altering systems throughout their **life cycles**. You will agree with me that, an effective System Development Life Cycle (SDLC) should result in a high-quality system that meets customer

expectations, reaches completion within time and cost evaluations, and works effectively and efficiently in the current and planned Information Technology infrastructure.

In-text Question
What is system Development Lifecycle?

Answer
System Development Life Cycle (SDLC) is a conceptual model which includes policies and procedures for developing or altering systems throughout their life cycles.

2.2 Phases of the System Development Lifecycle (SDLC)

Systems Development **Life Cycle** is a systematic approach which explicitly breaks down the work into phases that are required to implement either new or **modified Information System**.

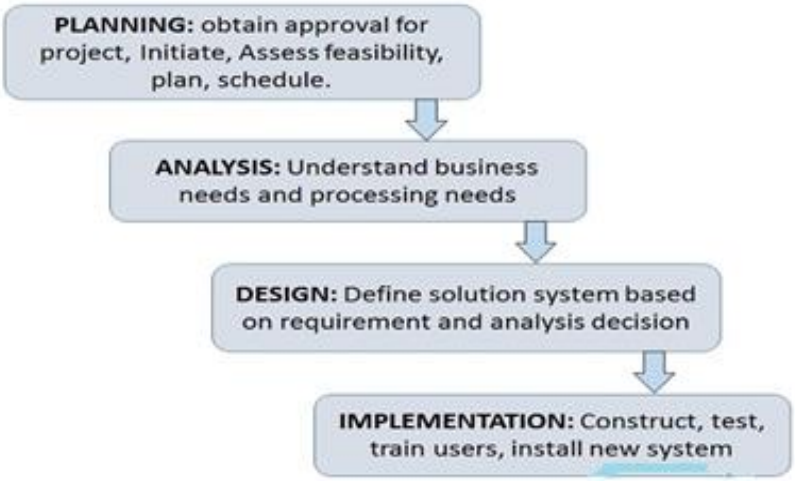

Fig 3.3.2: Phases of the system development lifecycle

2.2.1 Planning

You can define this as,the feasibility study phase, the planning phase encompasses the Definition of the problem and scope of existing system; Overview the new system and determine its objectives; Confirm project feasibility and produce the project Schedule; During this phase, you will observe that threats, constraints, integration and security of system are also considered; A feasibility report for the entire project is created at the end of this phase.

The planning phase is the fundamental process of understanding why an **information system** should be built and determining how the project team will go about building it. It has two steps:

1. During project initiation, the system's business value to the organisation is identified: How will it lower costs or increase revenues? Most ideas for new systems come from outside the IS area (e.g., **from the marketing department, accounting department**) in the form of a system request. A system request presents a brief summary of a business need, and it explains how a system that supports the need will create business value. The IS department works together with the person or department that generated the request (**called the project sponsor**) to conduct a feasibility analysis. The system request and feasibility analysis are presented to an information system approval committee (**sometimes called a steering committee**), which decides whether the project should be undertaken.

2. Once the project is approved, it enters project management. During project management, the project manager creates a work plan, staffs the project, and puts techniques in place to help the project team control and direct the project through the entire SDLC. The deliverable for project management is a project plan, which describes how the project team will go about developing the system.

2.2.2 Analysis Phase

This will sound interestingly to know that,the analysis phase answers the questions of who will use the **system**, what the system will do, and where and when it will be used. During this phase, we observe that the project team investigates any current system(s), identifies improvement opportunities, and develops a concept for the new **system**. The Analysis phase gathers, analyses, and validates the information about the system. At the phase also the requirements and prototypes for new **system** are defined; Evaluation of

alternatives and prioritization of the requirements is done; Examination of the information needs of **end-user and enhances** the system goal. A Software Requirement Specification (SRS) document, which specifies the software, hardware, functional, and network requirements of the system is prepared at the end of this phase.

This phase has three steps:

1. An ***analysis strategy***is developed to guide the project team's efforts. Such a strategy usually includes a study of the current system (called the *as-is system*) and its problems, and envisioning ways to design a new system (called the *to-be system*).

2. The next step is ***requirements gathering***(e.g., through interviews, group workshops, or questionnaires). The analysis of this information in conjunction with input from the project sponsor and many other people leads to the development of a concept for a new system. The system concept is then used as a basis to develop a set of business ***analysis models***that describes how the business will operate if the new system were developed. The set typically includes models that represent the data and processes necessary to support the underlying business process.

3. The analyses, system concept, and models are combined into a document called the ***system proposal***, which is presented to the project sponsor and other key decision makers (e.g., **members of the approval committee**) who will decide whether the project should continue to move forward. The system proposal is the initial deliverable that describes what business requirements the new system should meet. Because it is really the first step in the design of the new system, some experts argue that it is inappropriate to use the term *analysis* as the name for this phase; some argue a better name would be ***analysis and initial design***. Because most organizations continue to use the name *analysis* for this phase, we will use it in this book as well. It is important to remember,

however, that the deliverable from the analysis phase is both an analysis and a high-level initial design for the new system.

2.2.3 Design Phase

The ***design phase***decides *how* the system will operate, in terms of the hardware, software, and network infrastructure; the user interface, forms, and reports; and the specific programs, databases, and fi les that will be needed. Although most of the strategic decisions about the system were made in the development of the system concept during the analysis phase, the steps in the design phase determine exactly how the system will operate. The design phase has four steps:

1. The ***design strategy***is first developed. It clarifies whether the system will be developed by the company's own programmers, whether the system will be outsourced to another firm (**usually a consulting firm**), or whether the company will buy an existing software package.

2. This leads to the development of the basic *architecture design* for the system, which describes the hardware, software, and network infrastructure to be used. In most cases, the system will add or change the infrastructure that already exists in the organisation.

The *interface design* specifies how the users will move through the system (e.g., navigation methods such as menus and on-screen buttons) and the forms and reports that the system will use.

3. The ***database and file specifications***are developed. These define exactly what data will be stored and where they will be stored.

4. The analyst team develops the *program design*, which defines the programs that need to be written and exactly what each program will do.

This collection of deliverables (architecture design, interface design, database and file specifications, and program design) is the *system specification* that is handed to the programming team for implementation. At the end of the design

phase, the feasibility analysis and project plan arere-examined and revised, and another decision is made by the project sponsor and approvalcommittee about whether to terminate the project or continue.

Implementation

The final phase in the SDLC is the ***implementation phase***, during which the system is actuallybuilt (or purchased, in the case of a packaged software design). This is the phase that usually gets the most attention, because for most systems it is the longest and most expensive single part of the development process.

This phase has three steps:

1. System ***construction***is the first step. The system is built and tested to ensure that it performs as designed. Since the cost of fixing bugs can be immense, testing is one of the most critical steps in implementation. Most organisations spend more time and attention on testing than on writing the programs in the first place.

2. The system is installed. *Installation* is the process by which the old system is turned off and the new one is turned on. There are several approaches that may be used to convert from the old to the new system. One of the most important aspects of conversion is the *training plan*, used to teach users how to use the new system and help manage the changes caused by the new system.

3. The analyst team establishes a *support plan* for the system. This plan usually includes a formal or informal post-implementation review, as well as a systematic way for identifying major and minor changes needed for the system.

System Maintenance

System maintenance include all the activities such as phone support or physical on-site support for users that is required once the system is installing.Implement the changes that the **system might undergo** over a period of time, or implement any new requirements after the system is deployed at the customer location.It also includes handling the residual errors and resolve any issues that may exist in the system even after the testing phase.Maintenance and support may be needed for a longer time for **large systems** and for a short time for smaller systems.

In-text Question
What are the four phases of system development lifecycle?

Answer
Planning,
Analysis,
Design and implementation

3.0Conclusion/Summary

This study session has provided you with an understanding of system development lifecycle and its different phases of **system development**.

5.0Self-Assessment Questions

1. Briefly discuss the synergy between the different phases of system development.

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=mH-Nc5kvyQQ Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2txfToA and critique it in the discussion forum

6.0References/Further Readings

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STUDY SESSION 4

System Development Models

Section and Subsection Headings:

Introduction

1.0 Learning Outcomes

2.0 Main Content

2.1- Overview of System Development Models

2.1.1- Waterfall Model

2.1.2- Rapid Application Development (RAD)

2.1.3- AGILE

3.0 Study Session Summary and Conclusion

4.0 Self-Assessment Questions

5.0 Additional Activities (Videos, Animations & Out of Class activities)

6.0 References/Further Readings

Introduction:

You are welcome to study session 4, you will learn and explorethe different **system development models**.

1.0 Study Session Learning Outcomes

After you have studying this session, you should be able to

1. Discuss the different system development models.

2.0 Main Content

2.1 Overview of system development model

The term '**model**' is conveniently employed in a number of different ways. It is used as a noun implying a representation, as an adjective implying a degree of perfection, or as a verb implying to demonstrate or to show what something is

like. The most fundamental feature of models is that their construction has involved a highly selective attitude to information.

Models can be viewed as selective approximations, which, by the elimination of incidental detail, allow some fundamental, relevant or interesting aspects of the real world to appear in some, generalized form. Thus, models can be thought of as selective pictures and a direct description of the logical characteristics of our knowledge of the external world shows that each of these pictures gives undue prominence to some features of our knowledge and obscures and distorts the other features that rival pictures emphasize.

Fig 3.4.1: system development model

Each of them directs such a bright light on our part of the scene that it obscures other parts in a dark shadow. Only by being unfaithful in some respect can a model represent its original.

There are many different **systems development models**, and they vary in terms of the progression that is followed through the phases of the SDLC. Many organizations have their own **internal methodologies** that have been refined over the years, and they explain exactly how each phase of the SDLC is to be performed in that company.

Here we will review several of the predominant methodologies that have evolved over time.

2.1.1 Waterfall Model

The waterfall development model originated in the manufacturing and construction industries; where the highly structured physical environments meant that design changes became prohibitively expensive much sooner in the development process. When first adopted for software development, there were no recognized alternatives for knowledge-based creative work. With ***waterfall development***, analysts and users proceed sequentially from one phase to the next.

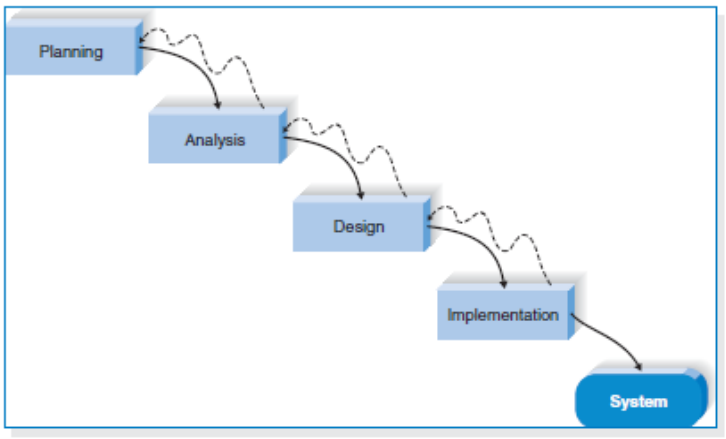

Fig 3.4.2: The Waterfall Model

There are two important variants of waterfall development.

The ***parallel development*** model evolved to address the lengthy time frame of waterfall development. As shown in fig 2.1.1b, instead of doing the design and implementation in sequence, a general design for the whole system is performed. Then the project is divided into a series of subprojects that can be designed and implemented in parallel. Once all subprojects are complete, there is a final integration of the separate pieces, and the **system** is delivered. Parallel development reduces the time required to deliver a system, so changes in the business environment are less likely to produce the need for rework.

The approach still suffers from problems caused by voluminous deliverables. It also adds a new problem: If the subprojects are not completely independent, design decisions in one subproject may affect another, and at the project end, integrating the subprojects may be quite challenging.

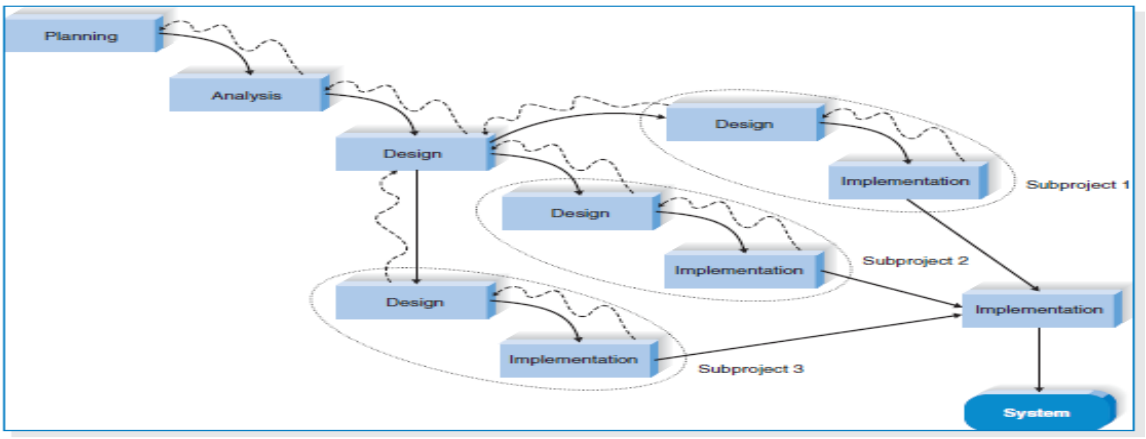

Fig 3.4.3: Parallel Model

The ***V-model*** is another variation of waterfall development that pays more explicit attention to testing. As shown in Figure 2.1.1c, the development process proceeds down the left-hand slope of the V, defining requirements and designing system components. At the base of the V, the code is written. On the upward-sloping right side of the model, testing of components, integration testing, and, finally, acceptance testing is performed. A key concept of this model is that as requirements are specified and components designed, testing for those elements is also defined. In this manner, each level of testing is clearly linked to a part of the analysis or design phase, helping to ensure high quality and relevant testing and maximize test effectiveness.

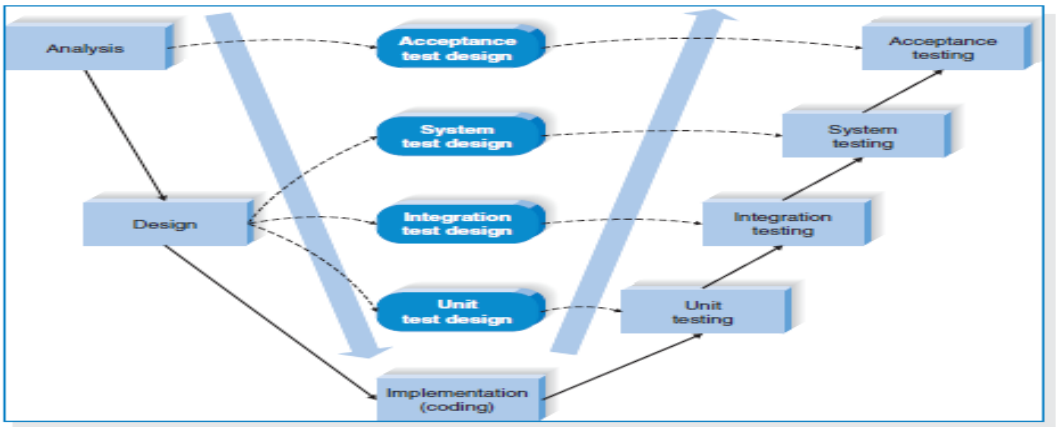

Fig 3.4.4: The V-Model

The V-model is simple and straightforward and improves the overall quality of systems through its emphasis on early development of test plans. Testing focus

and expertise is involved in the project earlier rather than later; plus, the testers gain knowledge of the project early. It still suffers from the rigidity of the waterfall development process, however, and is not always appropriate for the dynamic nature of the business environment.

2.1.2 Rapid Application Development Model

Rapid application development is a collection of models that emerged in response to the weaknesses of waterfall development and its variations. RAD incorporates special techniques and computer tools to speed up the analysis, design, and implementation phases in order to get some portion of the system developed quickly and into the hands of the users for evaluation and feedback. CASE (**computer-aided software engineering**) tools, JAD (joint application development) sessions, fourth-generation/visual programming languages (e.g., Visual Basic.NET), and code generators may all play a role in RAD. While RAD can improve the speed and quality of systems development, it may also introduce a problem in managing user expectations. As systems are developed more quickly and users gain a better understanding of information technology, user expectations may dramatically increase and system requirements may expand during the project (**sometimes known as scope creep or feature creep**).

RAD may be conducted in a variety of ways. **Iterative development** breaks the overall project into a series of versions that are developed sequentially. The most important and fundamental requirements are bundled into the first version of the system.

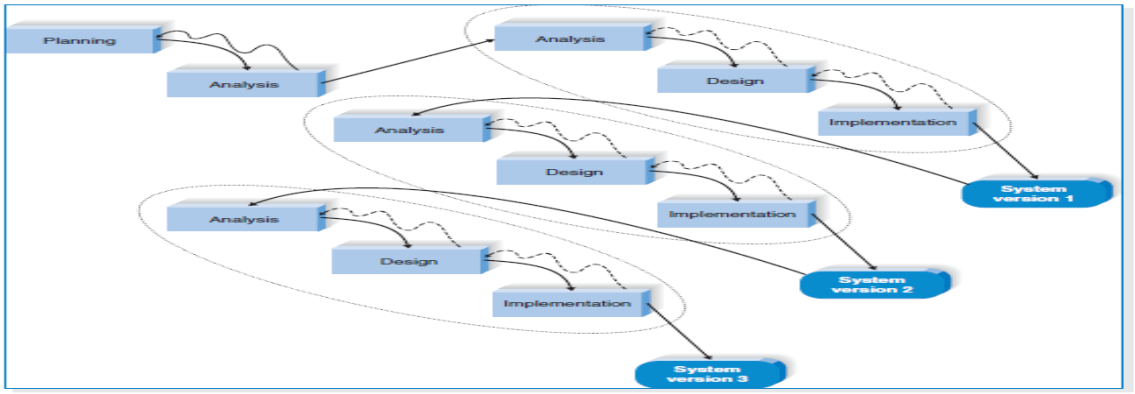

Fig 3.4.5: RAD Model.

2.1.3 AGILE Model

Agile SDLC model is a combination of iterative and **incremental process models** with focus on process adaptability and customer satisfaction by rapid delivery of working software product. Agile Methods break the product into small incremental builds. These builds are provided in iterations. Each iteration typically lasts from about one to three weeks. Every iteration involves cross functional teams working simultaneously on various areas like planning, requirements analysis, design, coding, unit testing, and acceptance testing.

Agile model believes that every project needs to be handled differently and the existing methods need to be tailored to best suit the project requirements. In agile the tasks are divided to time boxes small-time-frames to deliver specific features for a release. Iterative approach is taken and working software build is delivered after each iteration. Each build is incremental in terms of features; the final build holds all the features required by the customer.

In-text Question
List any three system development models?

Answer
Waterfall model, Rapid Application development model and AGILE Model

3.0Conclusion/Summary

This study session provided you with an understanding of system development

models.

4.0 Self-Assessment Questions

1. Briefly discuss any two system development models

5.0 Additional Activities (Videos, Animations &Out of Class activities) e.g.

a. Visit U-tube add https://www.youtube.com/watch?v=Y_A0E1ToC_I

for RAD: https://www.youtube.com/watch?v=yngKRepAgSg&t=31s

for AGILE: https://www.youtube.com/watch?v=cWQrJ7DGgXA&t=84s Watch the video & summarise in 1 paragraph

b. View the animation on add/sitehttps://bit.ly/2vFK9wS and critique it in the discussion forum

6.0 References/Further Readings

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XIII Glossary

ADC. analog-to-digital converter.

ALU. arithmetic logic unit.

ANSI. American National Standards Institute.

ASCII. American Standard Code for Information Interchange.

abstraction. The separation of the logical properties of data or function from its implementation in a computer program. See: encapsulation, information hiding, software engineering.

access. (ANSI) To obtain the use of a resource.

access time. (ISO) The time interval between the instant at which a call for data is initiated and the instant at which the delivery of the data is completed.

accident. See: mishap.

accuracy study processor. A software tool used to perform calculations or determine accuracy of computer manipulated program variables.

algorithm. (IEEE) (1) A finite set of well-defined rules for the solution of a problem in a finite number of steps. (2) Any sequence of operations for performing a specific task.

algorithm analysis. (IEEE) A software V&V task to ensure that the algorithms selected are correct, appropriate, and stable, and meet all accuracy, timing, and sizing requirements.

alphanumeric. Pertaining to a character set that contains letters, digits, and usually other characters such as punctuation marks.

auxiliary storage. Storage device other than main memory [RAM]; e.g., disks and tapes.

BIOS. basic input/output system.

bps. bits per second.

bandwidth. The transmission capacity of a computer channel, communications line or bus. It is expressed in cycles per second [Hz], and also is often stated in bits or bytes per second. See: band.

bar code. (ISO) A code representing characters by sets of parallel bars of varying thickness and separation that are read optically by transverse scanning.

baseline. (NIST) A specification or product that has been formally reviewed and agreed upon, that serves as the basis for further development, and that can be changed only through formal change control procedures.

binary. The base two number system. Permissible digits are "0" and "1".

bit. A contraction of the term binary digit. The bit is the basic unit of digital data. It may be in one of two states, logic 1 or logic 0. It may be thought of as a switch which is either on or off. Bits are usually combined into computer words of various sizes, such as the byte.

bits per second. A measure of the speed of data transfer in a communications system.

black-box testing. See: testing, functional.

block check. (ISO) The part of the error control procedure that is used for determining that a block of data is structured according to given rules.

block length. (1) (ISO) The number of records, words or characters in a block. (2) (ANSI) A measure of the size of a block, usually specified in units such as records, words, computer words, or characters.

block transfer. (ISO) The process, initiated by a single action, of transferring one or more blocks of data.

blocking factor. (ISO) The number of records in a block. The number is computed by dividing the size of the block by the size of each record contained therein. Syn: grouping factor.

blueprint. An exact or detailed plan or outline. Contrast with graph.

bootstrap. (IEEE) A short computer program that is permanently resident or easily loaded into a computer and whose execution brings a larger program, such an operating system or its loader, into memory.

branch. An instruction which causes program execution to jump to a new point in the program sequence, rather than execute the next instruction. Syn: jump.

branch analysis. (Myers) A test case identification technique which produces enough test cases such that each decision has a true and a false outcome at least once. Contrast with path analysis.

buffer. A device or storage area [memory] used to store data temporarily to compensate for differences in rates of data flow, time of occurrence of events, or amounts of data that can be handled by the devices or processes involved in the transfer or use of the data.

bug. A fault in a program which causes the program to perform in an unintended or unanticipated manner. See: anomaly, defect, error, exception, fault.

byte. A sequence of adjacent bits, usually eight, operated on as a unit.

cathode ray tube. An output device. Syn: display, monitor, screen.

cause effect graph. (Myers) A Boolean graph linking causes and effects. The graph is actually a digital-logic circuit (a combinatorial logic network) using a simpler notation than standard electronics notation.

change tracker. A software tool which documents all changes made to a program.

clock. (ISO) A device that generates periodic, accurately spaced signals used for such purposes as timing, regulation of the operations of a processor, or generation of interrupts.

coaxial cable. High-capacity cable used in communications and video transmissions. Provides a much higher bandwidth than twisted wire pair.

COBOL. Acronym for COmmon Business Oriented Language. A high-level programming language intended for use in the solution of problems in business data processing.

compatibility. (ANSI) The capability of a functional unit to meet the requirements of a specified interface.

compilation. (NIST) Translating a program expressed in a problem-oriented language or a procedure oriented language into object code. Contrast with assembling, interpret. See: compiler.

computer science. (ISO) The branch of science and technology that is concerned with methods and techniques relating to data processing performed by automatic means.

computerized system. Includes hardware, software, peripheral devices, personnel, and documentation; e.g., manuals and Standard Operating Procedures. See: computer, computer system.

configurable, off-the-shelf software. Application software, sometimes general purpose, written for a variety of industries or users in a manner that permits users to modify the program to meet their individual needs.

configuration audit. See: functional configuration audit, physical configuration audit.

data bus. (ANSI) A bus used to communicate data internally and externally to and from a processing unit or a storage device. See: bus.

data corruption. (ISO) A violation of data integrity. Syn: data contamination.

data exception. (IEEE) An exception that occurs when a program attempts to use or access data incorrectly.

data flow analysis. (IEEE) A software V&V task to ensure that the input and output data and their formats are properly defined, and that the data flows are correct.

data flow diagram. (IEEE) A diagram that depicts data sources, data sinks, data storage, and processes performed on data as nodes, and logical flow of data as links between the nodes. Syn: data flowchart, data flow graph.

data integrity. (IEEE) The degree to which a collection of data is complete, consistent, and accurate. Syn: data quality.

data item. (ANSI) A named component of a data element. Usually the smallest component.

data set. A collection of related records. Syn: file.

data sink. (IEEE) The equipment which accepts data signals after transmission.

data structure. (IEEE) A physical or logical relationship among data elements, designed to support specific data manipulation functions.

data structure centered design. A structured software design technique wherein the architecture of a system is derived from analysis of the structure of the data sets with which the system must deal.

database analysis. (IEEE) A software V&V task to ensure that the database structure and access methods are compatible with the logical design.

database security. The degree to which a database is protected from exposure to

default. (ANSI) Pertaining to an attribute, value, or option that is assumed when none is explicitly specified.

default value. A standard setting or state to be taken by the program if no alternate setting or state is initiated by the system or the user. A value assigned automatically if one is not given by the user.

defect. See: anomaly, bug, error, exception, fault.

defect analysis. See: failure analysis.

delimiter. (ANSI) A character used to indicate the beginning or the end of a character string. Syn: separator.

demodulate. Retrieve the information content from a modulated carrier wave; the reverse of modulate. Contrast with modulate.

demodulation. Converting signals from a wave form [analog] to pulse form [digital]. Contrast with modulation.

digital. Pertaining to data [signals] in the form of discrete [separate/pulse form] integral values. Contrast with analog.

digital-to-analog converter. Output related devices which translate a computer's digital outputs to the corresponding analog signals needed by an output device such as an actuator. Contrast with ADC [Analog-to-Digital Converter].

direct memory access. Specialized circuitry or a dedicated microprocessor that transfers data from memory to memory without using the CPU.

directed graph. (IEEE) A graph in which direction is implied in the internode connections. Syn: digraph.

disk. Circular rotating magnetic storage hardware. Disks can be hard [fixed] or flexible [removable] and different sizes.

disk drive. Hardware used to read from or write to a disk or diskette.

disk operating system. An operating system program; e.g., DR-DOS from Digital Research, MS-DOS from Microsoft Corp., OS/2 from IBM, PC-DOS from IBM, System-7 from Apple.

diskette. A floppy [flexible] disk.

electronic media. Hardware intended to store binary data; e.g., integrated circuit, magnetic tape, magnetic disk.

extended ASCII. The second half of the ACSII character set, 128 thru 255. The symbols are defined by IBM for the PC and by other vendors for proprietary use. It is non-standard ASCII. See: ASCII.

extended binary coded decimal interchange code. An eight bit code used to represent specific data characters in some computers; e.g., IBM mainframe computers.

extremal test data. (NBS) Test data that is at the extreme or boundary of the domain of an input variable or which produces results at the boundary of an output domain. See: testing, boundary value.

.

fail-safe. (IEEE) A system or component that automatically places itself in a safe operational mode in the event of a failure.

failure. (IEEE) The inability of a system or component to perform its required functions within specified performance requirements. See: bug, crash, exception, fault.

failure analysis. Determining the exact nature and location of a program error in order to fix the error, to identify and fix other similar errors, and to initiate corrective action to prevent future occurrences of this type of error. Contrast with debugging.

Failure Modes and Effects Analysis. (IEC) A method of reliability analysis intended to identify failures, at the basic component level, which have

significant consequences affecting the system performance in the application considered.

fiber optics. Communications systems that use optical fibers for transmission. See: optical fiber.

file maintenance. (ANSI) The activity of keeping a file up to date by adding, changing, or deleting data.

flag. (IEEE) A variable that is set to a prescribed state, often "true" or "false", based on the results of a process or the occurrence of a specified condition. Syn: indicator.

flat file. A data file that does not physically interconnect with or point to other files. Any relationship between two flat files is logical; e.g., matching account numbers.

floppy disk. See: diskette.

floppy disk drive. See: disk, disk drive.

functional decomposition. See: modular decomposition.

functional design. (IEEE) (1) The process of defining the working relationships among the components of a system. See: architectural design. (2) The result of the process in (1).

functional requirement. (IEEE) A requirement that specifies a function that a system or system component must be able to perform.

GB. gigabyte.

gigabyte. Approximately one billion bytes; precisely 230 or 1,073,741,824 bytes. See: kilobyte, megabyte.

half duplex. Transmissions [communications] which occur in only one direction at a time, but that direction can change.

handshake. An interlocked sequence of signals between connected components in which each component waits for the acknowledgement of its previous signal before proceeding with its action, such as data transfer.

hard copy. Printed, etc., output on paper.

hard disk drive. Hardware used to read from or write to a hard disk. See: disk, disk drive.

hard drive. Syn: hard disk drive.

hardware. (ISO) Physical equipment, as opposed to programs, procedures, rules, and associated documentation. Contrast with software.

interface requirement. (IEEE) A requirement that specifies an external item with which a system or system component must interact, or sets forth constraints on formats, timing, or other factors caused by such an interaction.

International Electrotechnical Commission. Geneva, Switzerland. An organization that sets standards for electronic products and components which are adopted by the safety standards agencies of many countries.

International Standards Organization. See: International Organization for Standardization.

interrupt analyzer. A software tool which analyzes potential conflicts in a system as a result of the occurrences of interrupts.

JCL. job control language.

job. (IEEE) A user-defined unit of work that is to be accomplished by a computer. For example, the compilation, loading, and execution of a computer program. See: job control language.

job control language. (IEEE) A language used to identify a sequence of jobs, describe their requirements to an operating system, and control their execution.

key. One or more characters, usually within a set of data, that contains information about the set, including its identification.

key element. (QA) An individual step in an critical control point of the manufacturing process.

kilobyte. Approximately one thousand bytes. This symbol is used to describe the size of computer memory or disk storage space. Because computers use a binary number system, a kilobyte is precisely 210 or 1024 bytes.

LAN. local area network.

LSI. large scale integration.

ladder logic. A graphical, problem oriented, programming language which replicates electronic switching blueprints.

language. See: programming language.

latent defect. See: bug, fault.

life cycle. See: software life cycle.

life cycle methodology. The use of any one of several structured methods to plan, design, implement, test. and operate a system from its conception to the termination of its use. See: waterfall model.

machine language. See: machine code.

macro. (IEEE) In software engineering, a predefined sequence of computer instructions that is inserted into a program, usually during assembly or compilation, at each place that its corresponding macroinstruction appears in the program.

macroinstruction. (IEEE) A source code instruction that is replaced by a predefined sequence of source instructions, usually in the same language as the rest of the program and usually during assembly or compilation.

main memory. A non-moving storage device utilizing one of a number of types of electronic circuitry to store information.

main program. (IEEE) A software component that is called by the operating system of a computer and that usually calls other software components. See: routine, subprogram.

mainframe. Term used to describe a large computer.

mean time to repair. A measure of reliability of a piece of repairable equipment, giving the average time between repairs.

measure. (IEEE) A quantitative assessment of the degree to which a software product or process possesses a given attribute.

measurable. Capable of being measured.

measurement. The process of determining the value of some quantity in terms of a standard unit.

medium scale integration. A classification of ICs [chips] based on their size as expressed by the number of circuits or logic gates they contain. An MSI IC contains 100 to 3,000 transistors.

megabit. Approximately one million bits. Precisely 1024 K bits, 220 bits, or 1,048,576 bits.

multi-tasking. (IEEE) A mode of operation in which two or more tasks are executed in an interleaved manner. Syn: parallel processing. See: multi-processing, multi-programming, time sharing.

multiple condition coverage. (Myers) A test coverage criteria which requires

n-channel MOS. A type of microelectronic circuit used for logic and memory chips.

network. (1) (ISO) An arrangement of nodes and interconnecting branches. (2) A system [transmission channels and supporting hardware and software] that connects several remotely located computers via telecommunications.

null. (IEEE) A value whose definition is to be supplied within the context of a specific operating system. This value is a representation of the set of no numbers or no value for the operating system in use.

null data. (IEEE) Data for which space is allocated but for which no value currently exists.

null string. (IEEE) A string containing no entries. Note: It is said that a null string has length zero.

Oracle. A relational database programming system incorporating the SQL programming language. A registered trademark of the Oracle Corp.

original equipment manufacturer. A manufacturer of computer hardware.

overflow. (ISO) In a calculator, the state in which the calculator is unable to accept or process the number of digits in the entry or in the result. See: arithmetic overflow.

overflow exception. (IEEE) An exception that occurs when the result of an arithmetic operation exceeds the size of the storage location designated to receive it.

parallel processing. See: multi-processing, multi- programming.

parameter. (IEEE) A constant, variable or expression that is used to pass values between software modules. Syn: argument.

parity bit. (ISO) A binary digit appended to a group of binary digits to make the sum of all the digits, including the appended binary digit, either odd or even, as predetermined.

parity check. (ISO) A redundancy check by which a recalculated parity bit is compared to the predetermined parity bit. Contrast with check summation, cyclic redundancy check [CRC].

Pascal. A high-level programming language designed to encourage structured programming practices.

password. (ISO) A character string that enables a user to have full or limited access to a system or to a set of data.

patch. (IEEE) A change made directly to an object program without reassembling or recompiling from the source program.

path. (IEEE) A sequence of instructions that may be performed in the execution of a computer program.

peripheral equipment. See: peripheral device.

personal computer. Synonymous with microcomputer, a computer that is functionally similar to large computers, but serves only one user.

physical configuration audit. (IEEE) An audit conducted to verify that a configuration item, as built, conforms to the technical documentation that defines it. See: functional configuration audit.

program mutation. (IEEE) A computer program that has been purposely altered from the intended version to evaluate the ability of program test cases to detect the alteration. See: testing, mutation.

programmable array logic. A programmable logic chip. See: programmable logic device.

programmable logic array. A programmable logic chip. See: programmable logic device.

programmable logic device. A logic chip that is programmed at the user's site. Contrast with PROM.

protection exception. (IEEE) An exception that occurs when a program attempts to write into a protected area in storage.

protocol. (ISO) A set of semantic and syntactic rules that determines the behavior of functional units in achieving communication.

risk. (IEEE) A measure of the probability and severity of undesired effects. Often taken as the simple product of probability and consequence.

risk assessment. (DOD) A comprehensive evaluation of the risk and its associated impact.

robustness. The degree to which a software system or component can function correctly in the presence of invalid inputs or stressful environmental conditions. See: software reliability.

routine. (IEEE) A subprogram that is called by other programs and subprograms. Note: This term is defined differently in various programming languages. See: module.

RS-232-C. An Electronic Industries Association (EIA) standard for connecting electronic equipment. Data is transmitted and received in serial format.

safety. (DOD) Freedom from those conditions that can cause death, injury, occupational illness, or damage to or loss of equipment or property, or damage to the environment.

safety critical. (DOD) A term applied to a condition, event, operation, process or item of whose proper recognition, control, performance or tolerance is essential to safe system operation or use; e.g., safety critical function, safety critical path, safety critical component.

safety critical computer software components. (DOD) Those computer software components and units whose errors can result in a potential hazard, or loss of predictability or control of a system.

security. See: computer system security.

TCP/IP. transmission control protocol/Internet protocol.

tape. Linear magnetic storage hardware, rolled onto a reel or cassette.

telecommunication system. The devices and functions relating to transmission of data between the central processing system and remotely located users.

terabyte. Approximately one trillion bytes; precisely 240 or 1,099,511,627,776 bytes. See: kilobyte, megabyte, gigabyte.

terminal. A device, usually equipped with a CRT display and keyboard, used to send and receive information to and from a computer via a communication channel.

test. (IEEE) An activity in which a system or component is executed under specified conditions, the results are observed or recorded and an evaluation is made of some aspect of the system or component.

testing, interface. (IEEE) Testing conducted to evaluate whether systems or components pass data and control correctly to one another. Contrast with testing, unit; testing, system. See: testing, integration.

testing, interphase. See: testing, interface.

testing, invalid case. A testing technique using erroneous [invalid, abnormal, or unexpected] input values or conditions. See: equivalence class partitioning.

testing, mutation. (IEEE) A testing methodology in which two or more program mutations are executed using the same test cases to evaluate the ability of the test cases to detect differences in the mutations.

testing, operational. (IEEE) Testing conducted to evaluate a system or component in its operational environment. Contrast with testing, development; testing, acceptance; See: testing, system.

testing, qualification. (IEEE) Formal testing, usually conducted by the developer for the consumer, to demonstrate that the software meets its specified requirements. See: testing, acceptance; testing, system.

testing, regression. (NIST) Rerunning test cases which a program has previously executed correctly in order to detect errors spawned by changes or corrections made during software development and maintenance.

testing, special case. A testing technique using input values that seem likely to cause program errors; e.g., "0", "1", NULL, empty string. See: error guessing.

testing, statement. (NIST) Testing to satisfy the criterion that each statement in a program be executed at least once during program testing. Syn: statement coverage. Contrast with testing, branch; testing, path; branch coverage; condition coverage; decision coverage; multiple condition coverage; path coverage.

timing and sizing analysis. (IEEE) Analysis of the safety implications of safety-critical requirements that relate to execution time, clock time, and memory allocation.

top-down design. Pertaining to design methodology that starts with the highest level of abstraction and proceeds through progressively lower levels. See: structured design.

unit. (IEEE) (1) A separately testable element specified in the design of a computer software element. (2) A logically separable part of a computer program. Syn: component, module.

UNIX. A multitasking, multiple-user (time-sharing) operating system developed at Bell Labs to create a favorable environment for programming research and development.

usability. (IEEE) The ease with which a user can learn to operate, prepare inputs for, and interpret outputs of a system or component.

user. (ANSI) Any person, organization, or functional unit that uses the services of an information processing system. See: end user.

user's guide. (ISO) Documentation that describes how to use a functional unit, and that may include description of the rights and responsibilities of the user, the owner, and the supplier of the unit. Syn: user manual, operator manual.

utility program. (ISO) A computer program in general support of the processes of a computer; e.g., a diagnostic program, a trace program, a sort program. Syn: service program. See: utility software.

utility software. (IEEE) Computer programs or routines designed to perform some general support function required by other application software, by the operating system, or by the system users. They perform general functions such as formatting electronic media, making copies of files, or deleting files.

valid. (1) Sound. (2) Well grounded on principles of evidence. (3) Able to withstand criticism or objection.

valid input. (NBS) Test data that lie within the domain of the function represented by the program.